

Exploring drivers and policy enablers of citizen engagement in renewable energy projects through explainable AI

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HIGHLIGHTS

- Developed SCI and EIM metrics to measure explainers' sensitivity.
- Employed XGBoost and Tree SHAP to provide robust explanations.
- Plans to adopt renewable technologies polarize participation into high or low levels.
- Belief in corporate-owned projects drives both high and low engagement extremes.
- Policy risk awareness motivates moderate participation, challenging deterrence views.

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ABSTRACT

Community Renewable Energy (CRE) projects play a critical role in the global clean energy transition by enabling citizens to collectively own and manage energy systems. Despite growing interest, the factors driving participation—and the extent of their influence—remain underexplored. Most prior studies conceptualize participation narrowly, overlook the relative importance of drivers, and rely on traditional statistical methods that struggle to capture complex behavioral patterns. Addressing these gaps requires approaches suited to high-dimensional data with nonlinear interactions. Machine Learning (ML) offers the flexibility to model such complexity, while explainable Artificial Intelligence (XAI) techniques provide transparency by interpreting predictions independently of the underlying algorithm—essential for generating policy-relevant insights. Using Australia as a case study, this research addresses three questions: whether ML models can accurately predict citizens' multi-role participation levels; how robust model-agnostic explanations are when applied to heterogeneous behavioral data; and what factors drive varying levels of engagement. We surveyed 875 residents on their participation across ten CRE roles—from investing to volunteering and advocacy—synthesizing responses into a composite involvement score categorized as low, medium, or high. Classification algorithms including eXtreme Gradient Boosting (XGBoost), Multi-layer Perceptron (MLP), and Keras-based Sequential models were paired with SHapley Additive exPlanations (SHAP)—a model-agnostic XAI method—using Tree SHAP, Kernel SHAP, and Deep SHAP variants for interpretation. To evaluate explanation robustness, we developed two novel metrics: the Stability Correlation Index (SCI) and the Explanation Integrity Metric (EIM), which introduce targeted perturbations to key features. XGBoost with Tree SHAP achieved the highest stability, with SCI above 99% and EIM near zero at moderate perturbation levels. Key findings reveal that plans to adopt renewable technologies polarize participation into high or low levels, while perceptions of corporate-owned projects act as a double-edged factor driving both engagement extremes. Increased awareness of policy risks motivates moderate participation, suggesting opportunities to leverage risk communication. These insights offer practical strategies for policymakers and project developers seeking to broaden and deepen citizen engagement in CRE initiatives.

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1. Introduction

The global push toward Renewable Energy (RE), driven by urgent climate concerns and ambitious international agreements, has gained unprecedented momentum in recent years. As nations worldwide set increasingly bold targets for reducing carbon emissions and transitioning to sustainable energy sources, the landscape of power generation is undergoing a profound transformation (Schmidt et al., 2017; Adua et al., 2021)—shifting away from centralized, fossil-fuel-based systems toward more decentralized, sustainable models. Among these decentralized solutions, Community Renewable Energy (CRE) projects—where citizens collectively own and manage energy systems—play a critical role in this transition (Abada et al., 2020; Moroni et al., 2019). They offer technical (IEA, 2023) and environmental benefits, and promote social acceptance of RE infrastructure (Westrom, 2020), energy democracy (Mey and Diesendorf, 2018), and community resilience (Norbu et al., 2021).

CRE projects support a wide range of citizen involvement, enabling individuals and communities to become active participants in the energy transition. Citizens can contribute financially as investors (Cohen et al., 2021), offer land for project sites or construction materials, assist as volunteers (Fischer et al., 2021; McLean et al., 2021), advocate for projects, or serve in formal governance roles (Broughel and Hampl, 2018). These diverse roles reflect the multidimensional nature of participation (Bauwens, 2019; Alipour et al., 2023)—but also reveal a persistent challenge: while some citizens engage across several roles, many remain uninvolved. Australia illustrates this gap starkly. Although approximately 87% of Australians support a moderate to fast energy transition (Dyer, 2024), formal community engagement remains remarkably low—a 2023 impact assessment found only 2600 active members across 55 community energy groups nationally (Community Power Agency, 2023). This support-engagement gap is further evidenced by the contrast between individual and collective action: while over 4 million Australian households have installed rooftop solar, community-scale projects contributed just 12 MW of generation compared to 3.2 GW added by individual rooftops in 2024 (Clean Energy Council, 2025). Similarly, although RE investment reached a record \$12.7 billion in 2024, community groups raised approximately \$74.5 million—less than 1% of total national investment (Community Power Agency, 2023; Clean Energy Council, 2025). Understanding what drives this variation between high public support and low collective participation is therefore critical for designing more inclusive and effective energy transitions.

Despite growing interest in CRE initiatives, there remains a significant research gap in understanding what drives individuals to engage—and to what extent—in these projects. While participation occurs across diverse roles, most existing studies rely on qualitative case studies, descriptive statistics, or linear models (Kriemeyer et al., 2021; Bamberg et al., 2015; Alipour et al., 2022) that simplify engagement as a binary or single-role outcome (Piselli et al., 2021; Kalkbrenner and Roosen, 2016). These approaches are limited in their ability to capture the multidimensional nature of participation, where individuals may contribute financially, operationally, socially, or strategically—and often in combination. Moreover, although prior work identifies potential drivers such as socioeconomic status, environmental concern, trust, or policy awareness, it rarely assesses their relative importance (Kalkbrenner and Roosen, 2016; Goedkoop et al., 2022). As a result, the current evidence base provides limited guidance for designing interventions that can meaningfully broaden and deepen citizen engagement.

Capturing the multidimensional nature of participation requires analytical approaches suited to the inherent complexity of citizen engagement data. The combination of multiple participation roles—financial, operational, social, and strategic—often held simultaneously by individuals, generates high-dimensional variable spaces that traditional methods struggle to model effectively (Ferreira et al., 2024; Taghikhah et al., 2022). Moreover, participation behaviors exhibit nonlinear relationships; for instance, the effect of income on investment willingness may plateau beyond a certain threshold, or social trust may only influence engagement when combined with sufficient environmental concern

(Tam and Chan, 2018). These interactions, compounded by non-normal distributions typical of behavioral data (e.g., skewed participation frequencies, zero-inflated engagement counts), demand methods capable of handling such complexity. Machine Learning (ML) models can capture these high-dimensional, nonlinear patterns, while Explainable Artificial Intelligence (XAI) model-agnostic explanation techniques—which interpret predictions independently of the underlying algorithm—offer the transparency essential for policy-relevant insights (Molnar, 2020; Mohammadi, 2025; Taghikhah et al., 2024, 2026, 2021). These considerations motivate three research questions:

RQ1: *Can machine learning models accurately classify citizens into distinct, multi-role CRE participation levels?* RQ2: *How robust are model-agnostic explanations when applied to heterogeneous, high-dimensional behavioral data?* RQ3: *What factors drive varying levels of citizen engagement in CRE?*

To address these questions, we designed a behaviorally grounded survey of 875 Australian residents, structured around Social Identity Theory (Tajfel, 1979), the Theory of Planned Behavior (Ajzen, 1991), and the Norm Activation Model (Schwartz, 1977). Respondents reported their participation across ten distinct CRE roles—including financial, operational, advisory, and community-facing contributions—which we synthesized into a single “involvement score.” This score enabled us to categorize participation into low, medium, and high levels, providing a robust target variable for prediction and interpretation.

Specifically, ensemble and deep learning models—including eXtreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016), Multi-layer Perceptron (MLP) (Pedregosa et al., 2011), and Keras-based Sequential model (Chollet and Chollet, 2021)—are employed to capture complex variable relationships without distributional related assumptions. However, predictive performance alone is insufficient for domains involving social trust and policy design. We therefore incorporate SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017)-based explanation methods—Tree SHAP (Lundberg et al., 2020), Kernel SHAP (Lundberg and Lee, 2017), and Deep SHAP (Lundberg et al., 2017)—to generate interpretable, model-agnostic insights into feature importance. To assess the robustness of these explanations, two novel metrics are introduced—namely, the Stability Correlation Index (SCI) and the Explanation Integrity Metric (EIM). These metrics apply feature-specific perturbations to SHAP-identified key variables, enabling systematic assessment of the stability and contextual integrity.

2. Background

2.1. Citizen engagement in CRE

While early research on citizen participation in RE or CRE initiatives has largely relied on qualitative and case-based methods (Petrovics et al., 2024), more recent work has begun to use survey-based approaches to capture broader behavioral patterns. These survey studies allow for structured data collection across larger populations, making it possible to identify and compare predictors of participation in a more generalizable way (Leknoi et al., 2022). To understand the current state of this literature, we reviewed a set of survey-based studies that examine citizen engagement in CRE across different countries and contexts. Table 1 synthesizes these studies and categorizes them by their aim, method, theoretical framework, participation scope (e.g., single-role vs. multi-role), and type of behavioral measurement. This structured review helps clarify where the field currently stands and where significant gaps remain—particularly regarding participation complexity, methodological depth, and model transparency.

Across this body of work, research aims vary widely—ranging from predicting citizen interest in CRE participation (Morgan and Canfield, 2021), to understanding motivations and feasibility (Wang et al., 2022; Piselli et al., 2021), to examining the role of community factors (Goedkoop et al., 2022). Despite this diversity, most studies share a common goal of identifying determinants that influence individuals’ willingness or intent to participate in such initiatives. These studies are

Table 1

Analyzing quantitative studies related to citizen's participation in CRE-related initiatives using survey.

Source	Aim	Sample size	Method	Relevant theories	Participation scope	Roles	Contribution	Country
(Bamberg et al., 2015)	Determinants of participation in community initiatives	811	SEM, Regression Analysis	Dual-Pathway Model, SIT	Single-role, Intention only	Participating	Theory	Germany, Austria
(Kalkbrenner and Roosen, 2016)	Willingness to participate in CRE	954	Regression, Mediation Analysis	TPB, SIT, Trust Theory	Single-role, Intention only	Participating	Theory, Policy	Germany
(Piselli et al., 2021)	Holistic experience and CRE feasibility	205	Dynamic Simulation	Extended TAM (partial)	Single-role, Intention only	Participating	Theory, Policy	Italy
(Morgan and Canfield, 2021)	Predict consumer interest in CRE	195	SEM	Value-Belief-Norm, Diffusion of Innovation	Single-role, Composite-index	Participating	Theory	United States
(Kriemeyer et al., 2021)	Motivations for local energy program	81	Mixed-methods (interviews, focus groups, regression)	Energy Cultures Framework	Dual-role, Observed-behavior	Adjusting usage, Participating in planing	Theory, Policy	United States
(Fischer et al., 2021)	Participation in RE cooperatives	4210	Ordered Probit, Descriptive Stats	Economic Preferences, Social Norms	Dual-role, Intention only	Investing, Volunteering	Theory, Policy	Germany
(Conradie et al., 2021)	Intent to participate in CRE	727	SEM	TPB	Single-role, Intention only	Participating	Theory, Policy	Belgium
(Wang et al., 2022)	Compare motivations for CRE participation	1058	Agglomerative clustering, SEM	Partial application of TPB	Single-role, Composite-index	Participating	Theory, Policy	China
(Goedkoop et al., 2022)	Community factors in CRE involvement	439	Logistic Regression, Proportional Odds	SIT (partial), TPB	Multi-role, Observed-behavior	Initiating, Supporting, Using	Theory, Policy	Netherlands
(Neves et al., 2024)	Tech factors in CRE	1602	Mixed-methods (qualitative interviews, SEM)	S-O-R Framework	Single-role, Intention only	Participating	Theory, Policy	Portugal, Germany, Greece, Italy
<i>This study</i>	<i>Participation level in CRE</i>	<i>875</i>	<i>Deep Learning, Ensemble Learning, XAI</i>	<i>TPB, SIT, Norm Activation Model</i>	<i>Multi-role, Composite-index</i>	<i>Investing, Volunteering, Contracted, Governing, Technical advisor, Cultural advisor, Engaging community, Supplying inputs</i>	<i>Theory, Method, Policy</i>	<i>Australia</i>

Note: SEM = Structural Equation Modeling; TAM = Technology Acceptance Model; TPB = Theory of Planned Behavior; S-O-R = Stimulus-Organism-Response; SIT = Social Identity Theory.

grounded in a variety of theoretical frameworks, including Theory of Planned Behavior (TPB), Social Identity Theory (SIT), Trust Theory, and Value-Belief-Norm (VBN) model. They span multiple countries, including the United States, China, Italy, Germany, Austria, Belgium, the Netherlands, Portugal, and Greece, providing a global perspective on the topic. The sample sizes in these studies range from 81 to 4210 participants, indicating considerable variation in research scale.

While these studies contribute meaningfully to theoretical development and policy insights, three important gaps emerge from the review. *First*, most studies define participation narrowly, often measuring only a single form of involvement—typically intention to invest or join a project—rather than capturing the diverse roles individuals may adopt. Only a few studies, such as (Goedkoop et al., 2022), acknowledge multi-role behaviors, yet even these tend to simplify the complexity of real-world engagement. This limited scope restricts our understanding of how different dimensions of participation—financial, advisory, technical, or community-facing—interact and vary across individuals (RQ1). *Second*, the field remains methodologically conservative. The majority of existing works employ classical statistical methods such as Structural Equation Modeling (SEM), regression analysis, or clustering. While these techniques offer theoretical clarity and ease of interpretation, they are often ill-suited to model the nonlinear, high-dimensional nature of behavioral survey data (Li et al., 2024; Vowels, 2023). Contributions are thus primarily theoretical or policy-oriented, with limited advancement in analytical frameworks. Most notably, research on CRE participation shows limited use of ML and XAI approaches, in contrast to other areas of energy research, where these methods have been widely adopted to predict energy production, optimize grid operations, and manage

storage systems (Nakabi and Toivanen, 2021; Alquthami et al., 2022; Nguyen et al., 2024) (RQ2). *Third*, although the literature collectively aims to identify determinants of participation, the evidence remains fragmented. None systematically assesses the relative importance of factors such as trust, environmental concern, or policy awareness—nor examines how these factors vary across different levels of engagement. This limits actionable guidance for targeted interventions (RQ3).

2.2. Perturbation-based evaluation metrics in XAI

Perturbation-based evaluation methods are commonly used in XAI to assess the robustness and stability of explanation techniques (Schlegel and Keim, 2023; Ghorbani et al., 2019; Buschmeier et al., 2023). These methods introduce small variations in input data—such as adding random noise to responses—and observe how much explanation values change in response. If explanations shift significantly due to minor perturbations, the model's interpretations are likely unstable and less trustworthy (Ivanovs et al., 2021; Nauta et al., 2023).

Several widely-used metrics exist for evaluating explanation stability, including cosine similarity (Chu et al., 2018), top-k intersection (Ghorbani et al., 2019), and rank order correlation (Boopathy et al., 2020). Cosine similarity quantifies orientation similarity between feature importance vectors but can overlook magnitude changes (Chu et al., 2018). Rank order correlation evaluates ranking consistency but does not capture scale changes (Moradi and Samwald, 2021; Cheng et al., 2021). The top-k intersection metric highlights agreement on top predictors but does not quantify overall explanation shifts.

When applied to behavioral survey data, these metrics face significant challenges (Fawzi et al., 2018; Taghikhah et al., 2021). Behavioral

data often involve high dimensionality due to numerous survey items and contain categorical and ordinal variables capturing qualitative aspects of behavior, attitudes, and perceptions. Standard numerical encoding may not preserve inherent relationships between categories, potentially yielding misleading importance scores. Moreover, behavioral data frequently exhibit multicollinearity due to overlapping survey content or underlying latent constructs, causing models to distribute importance among correlated features unpredictably. Metrics like cosine similarity and rank order correlation assume numerical, continuous, and independent features, making them unsuitable for behavioral data that violate these assumptions. They may fail to capture how minor perturbations, amplified by data inconsistencies, result in disproportionately large explanation changes (Kindermans et al., 2019).

These limitations reflect fundamental conceptual gaps in existing approaches. First, existing perturbation-based evaluations treat all features as equally worthy of perturbation, applying uniform noise across the input space regardless of whether a feature contributes meaningfully to predictions (Artelt et al., 2021; Batic et al., 2023). This approach conflates two distinct properties, including resilience to arbitrary noise and stability of meaningful explanations. A model may appear robust simply because perturbations to irrelevant features leave predictions unchanged—yet the same model could yield unstable attributions when influential features are perturbed. For behavioral survey data, where a small subset of attitudinal and contextual variables typically dominates predictive power while dozens of peripheral items contribute marginally, this distinction is critical.

Second, when stability metrics are computed by averaging across features or instances separately—rather than evaluating the full explanation structure jointly—they can mask instabilities through aggregation. A feature may maintain consistent average importance across the population even when individual attributions shift dramatically, or an individual's attribution profile may appear stable in aggregate even when specific feature contributions change unpredictably.

Third, existing metrics operate at the global level, collapsing stability assessments across all outcome classes into a single score (Alvarez-Melis and Jaakkola, 2018). In multi-class behavioral prediction—where different psychological mechanisms may drive different engagement levels—such aggregation masks precisely the class-specific instabilities that matter most for interpretation and intervention design. Without robust evaluation metrics addressing these conceptual mismatches, conclusions about which factors matter most for behavioral outcomes remain questionable (RQ2).

2.3. Contributions

Our study contributes to the literature in two key ways, each aligned with our research questions related to the methodological advancements in evaluating explanation stability (RQ2), and a novel application of ML and XAI to understand multi-role CRE participation (RQ1, RQ3).

First, we develop two novel metrics—the SCI and the EIM—to address the fundamental challenge of reliably evaluating SHAP-based explanations in noisy behavioral data. Our approach introduces three methodological innovations that distinguish it from prior work. Unlike existing methods applying uniform noise across all input features (Artelt et al., 2021; Batic et al., 2023), we implement selective perturbation that targets only the most influential features as identified by SHAP values. This ensures robustness is assessed where it matters for interpretation rather than being inflated by irrelevant features. Additionally, rather than computing stability through feature-wise or instance-wise averaging—which can mask instabilities through aggregation—our SCI metric treats the full instances-by-features explanation matrix as a structured object, enabling detection of instabilities that emerge from the interaction between population-level and feature-level patterns. Stability is computed separately for each outcome class rather than collapsing across classes into a single global score, revealing asymmetric robustness patterns critical for intervention design in multi-class behavioral prediction. Building upon the stability framework proposed

by (Rosenfeld, 2021), our metrics capture both the consistency of feature attribution under perturbation (SCI) and the magnitude of explanation shifts (EIM), providing complementary assessment of explanation robustness.

Second, we apply these methodological innovations to analyze CRE participation, marking a departure from traditional statistical approaches in this field. Our application combines ML techniques—specifically ensemble and deep learning methods—with SHAP-based interpretation to detect complex, nonlinear patterns in participation behavior. Rather than using conventional binary assessments of participation, we develop a composite involvement score spanning ten distinct roles, from financial investment to volunteering and advocacy. This multidimensional approach provides deeper insights into the complexity of citizen participation in community energy initiatives.

3. Methodology

3.1. Preparing CRE participation dataset

This study aims to provide empirical insights into the factors influencing citizen participation levels in CRE projects. Our methodological approach entails designing a survey based on behavioral theories (Taghikhah et al., 2021) such as SIT (Tajfel, 1979), TPB (Ajzen, 1991), and the Norm Activation Model (Schwartz, 1977). In total, the survey includes 11 constructs measured through 90 items distributed across 24 questions. The dependent variable captures *actual behavioral participation* rather than intention or willingness: respondents were asked “What role do you play in the community energy group/project?” to indicate roles they *currently occupy*. Response options span ten distinct participation modes: Investor (q_{045}) contributing financially to the project; Employed or Contracted (q_{046}) working in a paid or contracted capacity; Volunteer (q_{047}) assisting voluntarily, from key organizing to event setup; Decision-Making Group (q_{048}) participating in boards or committees that make project decisions; Occasional Expertise (q_{049}) providing expert advice or serving on advisory groups sporadically; Traditional Custodian Advisor (q_{050}) offering insights based on traditional and local knowledge; Community Engagement (q_{051} , q_{052}) communicating about the project with locals or discussing it with government and business stakeholders; Resource Provision (q_{053}) providing or leasing land or spaces for the project; Supply of Inputs (q_{054}) contributing additional resources like crop waste for bioenergy. This approach distinguishes realized engagement from hypothetical intention, aligning with established behavioral measures in participation research (Ekman and Amnå, 2012; Verba et al., 1995). By employing multiple-choice options rather than simple binary responses, the survey captures a more granular view of respondents' participation levels. The full list of survey items is available in Table B1, Appendix B. Placing these behavioral constructs within the real-world context is essential to design effective policies.

3.1.1. Data collection

In Australia, the CRE sector is still in its early stages of development, particularly compared to the more established community energy initiatives in Europe. There is no unified national policy framework specifically designed to support such initiatives, although some states and territories, such as Victoria and New South Wales, have implemented their own programs to encourage community energy development. This inconsistent policy landscape presents challenges for the widespread adoption and scaling of CRE projects in the country. While exact figures are not available, it is estimated that only a small proportion of Australian citizens (less than 5%) are currently engaged in CRE projects or similar forms of community energy investments. However, the potential for increased citizen participation is substantial, as highlighted by the 2023 community engagement review report conducted by the Australian Energy Infrastructure Commissioner (Australian Energy Infrastructure Commissioner, 2024).

In early 2023, we distributed the survey through multiple on-line channels associated with Geni.Energy and the Community Power Agency. This multi-channel approach, including social media platforms, direct emails, and community group engagement, was chosen to maximize accessibility and capture a wide range of perspectives. The online format facilitated efficient data management and allowed for real-time monitoring of response patterns, enabling us to adjust our sampling strategy as needed to maintain representativeness. We targeted Australian residents aged 18 and above, employing a stratified random sampling technique to ensure representation across various demographic and geographic segments. This approach was crucial given the potential regional variations in CRE engagement and attitudes toward RE.

The sample size was determined using the Cochran formula (Cochran, 1977), aiming for a 95% confidence level with a 5% margin of error. This rigorous approach to sample size calculation ensured statistical robustness for our subsequent analyses. Our data collection efforts yielded 875 complete responses. Detailed information on the data collection process is presented in Appendix A.

3.1.2. Dataset description

Appendix B presents the analysis of the dataset used in this study. Skewness and kurtosis values, combined with normality tests such as the Kolmogorov-Smirnov test, were calculated for each variable to assess the data adherence to normal distribution assumptions. For many variables, including “Environmental problems are frequently exaggerated” (q_{083}), “Personally, I feel responsible for climate change” (q_{086}), and “I trust the government to take positive action on pollution and climate change” (q_{088}), the skewness values were slightly negative, ranging from -0.5 to -0.69 , indicating a left-tailed distribution. Similarly, the kurtosis values were mostly close to 0, reflecting a mesokurtic distribution that closely aligns with normality.

The results of the Kolmogorov-Smirnov test, with p-values consistently below 0.05, indicated that none of the variables follows a normal distribution. For instance, the p-value for q_{083} was 0.00, confirming a significant deviation from normality. This pattern is common in behavioral survey data, where responses often cluster around neutral or extreme values. Skewness and kurtosis measures further support this: for example, the item “I consider myself well-informed about renewables and energy” (q_{084}) showed skewness of -0.30 and kurtosis of -0.37 , values relatively close to normal but still outside ideal thresholds. Conversely, items from “Renewables undermine the appearance of the landscape” (q_{119}) to “Profits from renewables go to external investors” (q_{116}) displayed mild positive skewness (at 0.04 and 0.02, respectively), suggesting a slight clustering of responses toward higher values. The findings indicated that most variables in the dataset do not follow a perfect normal distribution, which is expected in survey-based research. The lack of normality across features was addressed in the model development phase by utilizing ML algorithms that are robust to non-normal distributions.

3.1.3. Involvement score

To quantify participation levels in CRE projects, we developed a weighted scoring system based on the roles reported by the respondents. Let r_{pj} be an indicator equal to 1 if respondent p reported role j , and w_j denote the weight of role j , assigned based on domain knowledge. In this study, we used a straightforward approach, in which each role is assigned an equal weight of 1, providing a solid foundation for understanding the range of citizen engagement.

The involvement score of respondent p , S_p , is computed as in Eq. 1:

$$S_p = \sum_{j=1}^{10} r_{pj} w_j \quad \text{where } r_{pj} \in \{0, 1\}, \quad w_j \in \mathbb{R}^+ \quad (1)$$

where the summation covers 10 distinct roles ($j = 1, 2, \dots, 10$) identified in our survey. The distribution of S_p was analyzed using visual and statistical methods to determine appropriate categorization thresholds. We

examined the frequency distribution and box plot of the scores, focusing on measures of central tendency and spread, including the median, interquartile range, and notable peaks in the distribution.

Based on this analysis, we developed a three-category classification system for participation levels as in Eq. 2:

$$\text{Low} : 0 \leq S_p \leq a, \quad \text{Medium} : a < S_p \leq b, \quad \text{High} : S_p > b \quad (2)$$

where a and b are threshold values derived from distribution analysis. The Low Participation category was defined as scores below or equal to a , representing the more frequently occurring lower scores. The Medium Participation category spanned scores between a and b , capturing the central tendency of the data. The High Participation category encompassed scores above b , indicative of more extensive involvement in CRE projects. The resulting categories serve as the target set L for our subsequent model classifications.

For our predictive modeling, we define Q as the set of survey variables, excluding those directly related to roles in CRE projects shown in Eq. 3:

$$Q = \{q_1, q_2, \dots, q_m\} \quad (3)$$

where q_i represents the i -th survey question and m is the total number of questions excluding role-related items. These variables serve as the independent variables or predictors in our analysis. To ensure compatibility with our ML algorithms, we applied one-hot encoding to all categorical variables within Q . For a categorical variable q_k with n_k possible values, this encoding creates n_k binary variables as in Eq. 4:

$$q_k \rightarrow \{q_{k1}, q_{k2}, \dots, q_{kn_k}\} \quad \text{where } q_{ki} \in \{0, 1\}, \quad \sum_{i=1}^{n_k} q_{ki} = 1 \quad (4)$$

Note that q_{ki} is equal to 1 if q_k takes on the i -th category, and 0 otherwise.

3.2. Adopting ML methods for predicting and explaining CRE participation level

3.2.1. Ensemble and deep learning methods for prediction

The models built in this study aim to predict the levels of participation in CRE projects for each respondent p . Let $X \in \mathbb{R}^{N \times M}$ denote the matrix of predictor variables derived from Q , where N is the total number of instances (respondents) and M is the total number of features (predictors). Each row $X_p \in \mathbb{R}^M$ represents the responses of the p -th individual. Let $\ell \in \{\text{Low}, \text{Medium}, \text{High}\}^N$ be the vector of target variables, where each element ℓ_p represents the participation level of respondent p .

Formally, we define the classification problem as follows:

Given :

A feature matrix $X \in \mathbb{R}^{N \times M}$,

where each row $X_p \in \mathbb{R}^M$ corresponds to the p -th respondent,

and each column corresponds to a survey question.

A target vector $\ell \in \{\text{Low}, \text{Medium}, \text{High}\}^N$,

where ℓ_p denotes the participation level of respondent p .

Find :

$\hat{f} : \mathbb{R}^M \rightarrow \{\text{Low}, \text{Medium}, \text{High}\}$,

a classifier that maps X_p to the predicted label $\hat{\ell}_p$.

The objective is to learn a classifier $\hat{f}$ that accurately predicts the participation level ℓ_p for a given respondent p based on their predictor variables X_p . In developing predictive models that are robust to non-normality of features in the dataset, we employed XGBoost (Chen and Guestrin, 2016), Scikit-learn MLP (Pedregosa et al., 2011), and Keras-based Sequential (Chollet and Chollet, 2021) classifiers for the modeling. The train and test sets were primarily stratified to maintain an approximately equal distribution of class labels as in the original dataset. The hyperparameters of the classifiers were optimised using Grid Search (for

XGBoost and MLP), which exhaustively searches through a specified grid to maximize the model score (Pedregosa et al., 2011) and Keras Random Search Tuner (for Sequential) (O'Malley et al., 2019), which efficiently selects optimal combinations of hyperparameters from a randomly sampled space. To improve prediction performance of the minority class (High), class weights were incorporated into each tuning process, which was refined using 5-fold Cross Validation. Subsequently, each fine-tuned classifier is fitted to the survey data and evaluated.

The prediction performance of the trained classifiers was evaluated using accuracy, weighted F1 score and the Area Under Curve of the macro Receiver Operating Characteristic (ROC-AUC). Accuracy measures the proportion of correct predictions (both true positives and true negatives) out of the total number of cases examined. It provides a general idea of how often the classifier is correct across all classes. The F1 score is the harmonic mean of precision and recall, offering a balance between them. The weighted F1 score adjusts the F1 score for each class by the number of true instances for each class. ROC-AUC metric evaluates the performance across all classification thresholds. The ROC curve plots the true positive rate (sensitivity) against the false positive rate (1-specificity) at various threshold settings. The AUC measures the entire two-dimensional area underneath the entire ROC curve from (0,0) to (1,1). The macro-average computes this metric independently for each class and then takes the average, hence treating all classes equally regardless of their frequency. By incorporating ROC-AUC, we ensure that the model's performance is not biased by the majority class dominance, thereby giving a more balanced view of its effectiveness across all classes.

3.2.2. SHAP-based methods for model explanation

The primary goal of XAI is to provide explanations for how an ML system makes decisions or generates predictions, rather than treating the system as a black box (Ghaffarian et al., 2023; Maier et al., 2024; Taghikhah et al., 2022). One fundamental technique is SHAP (Lundberg and Lee, 2017), which has been integrated substantially with other techniques such as Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016) and Deep Learning Important Features (DeepLIFT) (Shrikumar et al., 2016). These methods collectively enhance the capability of XAI to provide advanced and comprehensible explanations for complex ML models. In this study, we applied Tree SHAP (Lundberg et al., 2020) to XGBoost, Kernel SHAP to MLPClassifier, and Deep SHAP (Lundberg et al., 2017) to Keras-based Sequential model. We fitted Tree SHAP to XGBoost as the explainer is tailored for tree-based models and can efficiently provide explanations for XGBoost which builds an ensemble of decision trees. On the other hand, Kernel SHAP can provide model-independent explanations by perturbing input features and observing changes in model output, making it effective for complex, non-linear models, and therefore was selected to explain MLPClassifier. Deep SHAP was utilised for Keras-based Sequential model, since the explainer leverages the internal structure of deep learning neural networks, providing accurate and efficient explanations. The mathematical foundations for these algorithms are provided in the Appendix C.

3.3. Developing perturbation-based evaluation metrics for behavioral data

In this study, we develop two perturbation-based evaluation metrics—the SCI and the EIM—integrated within a Selective Noise Addition framework (Algorithms 1–2) that targets only the most influential features as identified by SHAP values. Unlike existing approaches that apply uniform noise across all input features (Artelt et al., 2021; Batic et al., 2023), our framework restricts perturbation to features that actually drive predictions, ensuring robustness is assessed where it matters most for interpretation. SCI employs the Pearson Correlation Coefficient to measure the similarity between original and perturbed SHAP-based explanations, but applies it across the flattened explanation tensor of dimensions $N \times M$ rather than between simple paired vectors. This formulation captures how well feature importance rankings

are preserved after introducing perturbations—not for individual features or instances in isolation, but across the entire explanation space simultaneously. In simpler terms, it shows whether the relative importance structure remains stable even when targeted changes are made to influential variables. EIM adopts Mean Absolute Error (MAE) to quantify explanation drift, but differs from standard MAE applications that compare predictions against ground truth. Instead, EIM measures the average deviation between two explanation states—original and perturbed—across the full $N \times M$ tensor, normalized to provide a scale-independent measure of how much the explanation surface shifts. While correlation-based measures can indicate preserved rankings even when attribution magnitudes change substantially, EIM detects such scale shifts directly.

The SCI and EIM are defined as follows:

For a multi-class classification problem, let $S_{\text{orig}} \in \mathbb{R}^{C \times N \times M}$ represent the tensor of original SHAP values, and $S_{\text{noisy}} \in \mathbb{R}^{C \times N \times M}$ represent the tensor of SHAP values after perturbation. Here, C denotes the total number of classes.

3.3.1. Stability correlation index (SCI)

The SCI for each class c is defined in Eq. 5:

$$SCI^c = \frac{\sum_{i=1}^N \sum_{j=1}^M (S_{\text{orig},i,j}^c - \bar{S}_{\text{orig}}^c)(S_{\text{noisy},i,j}^c - \bar{S}_{\text{noisy}}^c)}{\sqrt{\sum_{i=1}^N \sum_{j=1}^M (S_{\text{orig},i,j}^c - \bar{S}_{\text{orig}}^c)^2} \sqrt{\sum_{i=1}^N \sum_{j=1}^M (S_{\text{noisy},i,j}^c - \bar{S}_{\text{noisy}}^c)^2}} \quad (5)$$

where $S_{\text{orig},i,j}^c$ is the SHAP value for class c , instance i , and feature j in the original dataset, $S_{\text{noisy},i,j}^c$ is the corresponding SHAP value for class c , instance i , and feature j in the perturbed dataset, $\bar{S}_{\text{orig}}^c$ is the mean of the original SHAP values for class c , and $\bar{S}_{\text{noisy}}^c$ is the mean of the perturbed SHAP values for class c . This formulation enables detection of class-specific instabilities invisible to existing metrics. A high SCI^c indicates that the feature importance rankings for class c remain stable even after data perturbations, demonstrating the robustness of the explainer.

Although SCI employs a correlation-based formulation, it differs from standard Pearson correlation in structure. Standard Pearson operates on two vectors of length n , measuring linear association between paired observations. SCI instead treats the full SHAP matrix (instances $\times$ features) as a single structured object, evaluating covariance across the joint space as shown in Eq. 6:

$$\text{Cov}(S_{\text{orig}}, S_{\text{noisy}}) = \frac{1}{N \cdot M} \sum_{i=1}^N \sum_{j=1}^M (S_{\text{orig},i,j}^c - \bar{S}_{\text{orig}}^c)(S_{\text{noisy},i,j}^c - \bar{S}_{\text{noisy}}^c) \quad (6)$$

This joint formulation differs from decomposed approaches, in which correlation is computed separately for each feature (feature-wise) or each instance (instance-wise). Such decomposed evaluations assess stability in isolated dimensions and may fail to detect instabilities that emerge from interactions between features and instances. By treating the full $N \times M$ matrix as a structured object, SCI provides holistic assessment of whether the explanation surface remains stable under perturbation.

3.3.2. Explanation integrity metric (EIM)

The EIM for each class c is defined in Eq. 7:

$$EIM^c = \frac{1}{N \cdot M} \sum_{i=1}^N \sum_{j=1}^M |S_{\text{orig},i,j}^c - S_{\text{noisy},i,j}^c| \quad (7)$$

This provides a direct measure of how much the explanations are affected by data alterations, with a lower EIM^c indicating higher robustness for class c . Standard MAE compares model predictions against ground truth labels—a single vector comparison where “error” implies deviation from a known correct answer. EIM instead quantifies drift between two explanation states, neither of which constitutes ground truth. The normalization by $N \times M$ ensures scale-independence across datasets

Algorithm 1 perturb_important_features.

Require: Test Set $\mathbf{X} \in \mathbb{R}^{N \times M}$, noise level $\epsilon \in \text{noise_levels}$, random seed $r \in \text{random_seeds}$, importance threshold k , SHAP explainer f .

Ensure: SCI_scores, EIM_scores

- 1: Compute original SHAP values S_{orig} for $\mathbf{X}$ using f
- 2: Compute mean absolute SHAP values $M_j = \frac{1}{m} \sum_{i=1}^m |s_{ij}| \quad \forall j \in \{1, 2, \dots, n\}$
- 3: Sort M_j in descending order of importance
- 4: Compute cumulative contribution $C_j = \sum_{k=1}^j M_k \quad \forall j \in \{1, 2, \dots, n\}$
- 5: Compute rate of change $R_j = \frac{C_j - C_{j-1}}{\max(C_{j-1}, \delta)} \quad \forall j \in \{2, \dots, n\}$ with $R_1 = 1$
- 6: Identify important features $\mathcal{I} = \{j \mid R_j > k\}$
- 7: Initialise SCI_scores and EIM_scores as empty lists of lists
- 8: **for all** $\epsilon \in \text{noise_levels}$ **do**
- 9: Initialise temporary lists of lists, SCI_temp_list and EIM_temp_list
- 10: **for all** $r \in \text{random_seeds}$ **do**
- 11: Add noise to important features: $\mathbf{X}_{\text{noisy}} = \text{add_feature_noise}(\mathbf{X}, \epsilon, \mathcal{I}, r)$ using Algorithm 2
- 12: Compute perturbed SHAP values S_{noisy} for $\mathbf{X}_{\text{noisy}}$
- 13: **for all** class index $c \in \text{class_indices}$ **do**
- 14: Compute scores SCI^c and EIM^c using S_{orig} and S_{noisy}
- 15: Append scores to SCI_temp_list[c], EIM_temp_list[c]
- 16: **end for**
- 17: **end for**
- 18: **for all** class index $c \in \text{class_indices}$ **do**
- 19: Compute averages, $\text{mean}(\text{SCI_temp_list}[c])$ and $\text{mean}(\text{EIM_temp_list}[c])$
- 20: Append averages to SCI_scores and EIM_scores
- 21: **end for**
- 22: **end for**
- 23: **return** SCI_scores, EIM_scores

Algorithm 2 add_feature_noise.

Require: Dataset $\mathbf{X} \in \mathbb{R}^{N \times M}$, noise level ϵ , important features $\mathcal{I}$, random seed r

Ensure: Noisy dataset $\mathbf{X}_{\text{noisy}}$

- 1: **if** $r \neq \text{None}$ **then**
- 2: Set the random seed using r
- 3: **end if**
- 4: Initialise $\mathbf{X}_{\text{noisy}}$ as a copy of $\mathbf{X}$
- 5: **for all** feature index $i \in \mathcal{I}$ **do**
- 6: Compute standard deviation σ_i of feature i
- 7: Generate noise $\mathbf{N}_i \sim \mathcal{N}(0, (\epsilon \cdot \sigma_i)^2)$
- 8: Add noise to feature i : $\mathbf{X}_{\text{noisy}}[:, i] = \mathbf{X}[:, i] + \mathbf{N}_i$
- 9: **end for**
- 10: **return** $\mathbf{X}_{\text{noisy}}$

of varying sizes. Eq. 8 captures the expected magnitude of attribution shift per instance-feature pair:

$$\mathbb{E}[|S_{\text{orig}} - S_{\text{noisy}}|] = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M |S^{\text{orig}}_{i,j} - S^{\text{noisy}}_{i,j}| \quad (8)$$

This formulation identifies magnitude instabilities overlooked by correlation-based metrics, where two explanation sets may be perfectly correlated but differ substantially in scale when all attributions shift proportionally.

To properly assess the robustness of SHAP explanations, we use the Selective Noise Addition Based on Feature Importance approach composed of two interconnected algorithms. Algorithm 1 identifies and perturbs important features, and Algorithm 2 systematically adds noise.

Feature selection is critical in the robustness testing of SHAP explainers, particularly when introducing perturbations to evaluate stability.

Algorithm 1 specifically targets the most influential features—those that have the greatest impact on the model predictions. By focusing on these key features, the algorithm makes the testing process both efficient and representative. This is especially important for high-dimensional behavioral datasets, where evaluating every feature would be impractical and less informative.

To ensure that the perturbations accurately reflect the natural variability of the data, Algorithm 2 scales the noise added to each feature based on its standard deviation. This approach ensures that the noise impact is equitably distributed across all features, regardless of their scale or range. By maintaining consistency in how features are affected by noise, the method facilitates a fairer and more accurate assessment of the explainer robustness. This methodological strategy allows us to better evaluate whether the SHAP explainer remains reliable when faced with different data conditions, thus enhancing the credibility and reliability of the model interpretations.

4. Results

We first presented how different ML models predict citizen participation in CRE projects and how well these predictions can be explained using interpretable methods. We then identified key factors influencing participation. All the analyses were conducted using Python, and the codes are available on GitHub at <https://github.com/Rosa-Th/SCI-EIM-ExplainableAI>.

4.1. Predictive model evaluation

The distribution of involvement scores (S_p), our target variable, is illustrated in Fig. 1. The interquartile range, which encompasses the middle 50% of S_p , spans from 2 to 3, with a median score of 2.5 and a notable peak at 3 in the frequency distribution. Consequently, we categorized the involvement scores of 0–2 as Low Participation (below the median and more frequent), 3 as Medium Participation (representing the central tendency), and ≥ 4 as High Participation (less frequent and indicative of higher involvement). The resulting categories yielded the following distribution of instances: Low Participation: 412 instances, Medium Participation: 405 instances, and High Participation: 58 instances.

Table 2 shows the results of fine-tuned hyperparameters of each optimal model fitted for explanations. Fig. 2 displays the comparative results of the evaluated models to predict the involvement scores for citizens, as presented by low, medium, and high participation classes. XGBoost exhibits superior performance metrics, achieving an accuracy and a weighted F1 score of 68%, with a macro ROC-AUC exceeding 81%. The Keras-based Sequential model demonstrates comparable performance, although with slightly lower metrics, while the MLP classifier exhibits the least favorable performance.

The individual ROC curves and confusion matrices were also examined to provide additional quantitative visualizations, as shown in Fig. 3. All models demonstrate a relatively strong ability to predict the minority ‘High’ class, as indicated by the highest ROC-AUC values (91% for XGBoost and Keras-based Sequential, 78% for MLP) among all classes. This higher ROC-AUC likely results from the models’ ability to confidently distinguish ‘High’ instances from the rest across all possible decision thresholds, even though they predict fewer ‘High’ true positives overall. The consistently high ROC-AUC values observed for the ‘High’ class across all classifiers can likely be attributed to the distinct feature patterns associated with high levels of engagement.

In contrast, while the models make more true positive predictions for the ‘Medium’ and ‘Low’ classes of involvement, these predictions might involve more ‘guessing’ rather than clear distinction, leading to lower ROC-AUC values for these classes. This suggests that the models are better at identifying ‘High’ instances across all possible decision thresholds, while their performance for ‘Medium’ and ‘Low’ classes is less consistent.

Furthermore, the XGBoost and Keras-based Sequential classifiers exhibit more robust class-wise predictions than the MLP, as aligned with their higher accuracies and weighted F1 scores. This indicates that

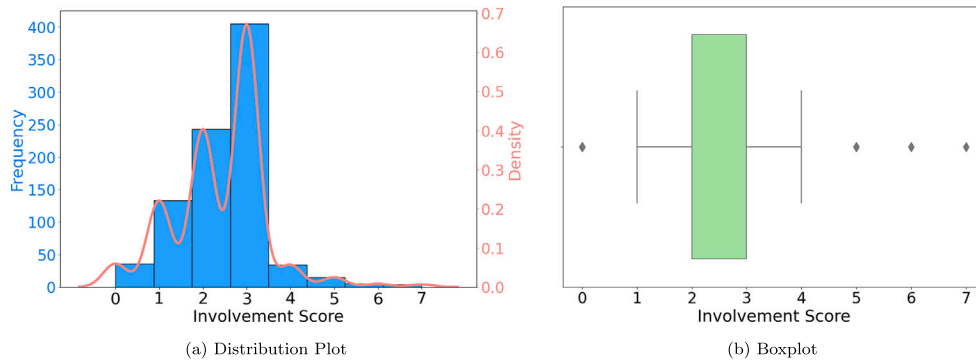


Fig. 1. Side-by-side visualization of involvement score distribution and boxplot.

Table 2

Fine-tuned hyperparameters for XGBoost, MLP, and Keras-based sequential models.

Model	Hyperparameters
XGBoost	Objective: multi:softprob, Max Depth: 4, Learning Rate: 0.1, Estimators: 50, Subsample: 0.8, Colsample by Tree: 0.7, Min Child Weight: 3, Gamma: 0
MLP	Activation: relu, Hidden Layers: (100, 100, 50), Learning Rate: 0.001, Alpha: 0.05, Batch Size: auto, Solver: adam, Max Iterations: 5000, Momentum: 0.9
Keras-based Sequential	Layers: 4, Units per Layer: 192, 32, 32, 32, Activation: tanh, relu, relu, relu, Learning Rate: 0.00032, Optimizer: rmsprop, Dropout: 0.2, 0.0, 0.0, 0.0, L1 Regularizer: 0.00205, 0.0001, 0.0001, 0.0001

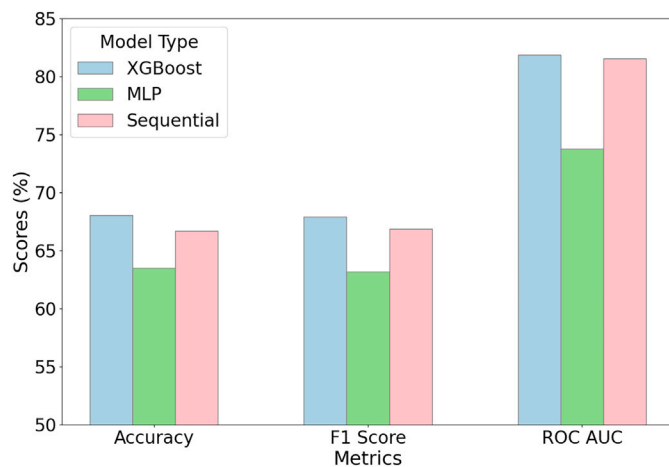


Fig. 2. Comparison of XGboost, MLP, and Keras-based Sequential models performance according to Accuracy, F1 Score and ROC-AUC Metrics.

XGBoost and Keras-based Sequential are generally better at distinguishing between the classes compared to MLP.

4.2. SHAP-based explainers evaluation

To establish a criterion for significance, we initially examined the rate of change in the cumulative feature contribution for each explainer, as illustrated in Fig. 4.

The rate of change represents how much each additional feature contributes to the overall explanatory power of the model, with sharp

declines indicating diminishing returns in terms of added interpretability. This helps identify the most influential features, beyond which the contributions of the remaining features are relatively insignificant. We used a 3% rate of change as a reasonable cutoff point, as values beyond this threshold showed a relatively flat trend, indicating less significant contributions. Using this cutoff, we identified 20, 23, and 25 important features out of a total of 91 features for the Tree, Kernel, and Deep explainers, respectively. Subsequently, we generated the corresponding metrics for each explainer perturbing these features. The noise levels were established at {0.005, 0.01, 0.02, 0.05, 0.1, 0.2}. The chosen levels cover a broad range from very small perturbations (0.5%) to relatively large perturbations (20%) of the feature values. This comprehensive range allows for evaluating model behavior under both minor and significant noise, providing a holistic view of the models robustness. For each level, random seeds from {0, 42, 150, 500, 999} were applied to accurately assess the average effects of perturbations.

Each explainer's performance was then evaluated using SCI scores at different perturbation levels and compared between the entire feature space and the important feature subset. Fig. 5 shows similar class-wise SCI patterns for Tree SHAP and Kernel SHAP, with the important feature subset yielding slightly higher SCI scores overall. This demonstrates that our approach reduces computational complexity and enhances the relevance of perturbation by focusing on the most important features, in line with domain knowledge that certain features are expected to have a more substantial impact on predictions and their explanations. Specifically, Tree SHAP is the most robust against perturbations, maintaining correlation scores above 99.5% for all classes at perturbation levels below 10%. Kernel SHAP follows closely, maintaining correlation above 95% for all classes at perturbation levels up to 5%. However, the Deep Explainer exhibited a less consistent pattern, with correlations fluctuating unpredictably against noise. For instance, the correlation for the Low class drops to approximately 1.1%, and for the Medium and High classes, it declines to -1.7% and -2.6% , respectively, even with a minimal 0.5% perturbation level, showing random variability across different perturbation levels. Appendix D.1 provides additional comparisons with standard Pearson correlation and decomposed averaging approaches, demonstrating that SCI detects instabilities masked by global aggregation.

Furthermore, the robustness of explanations provided by each explainer was assessed through EIM. The results, depicted in Fig. 6, indicate that the class-wise EIM scores of Tree SHAP and Kernel SHAP are slightly lower for the important feature subset compared to the entire feature set. This highlights the effectiveness of focusing on significant features during perturbation analysis. The EIM scores exhibit a pattern that is the inverse of the SCI scores, which is consistent with the expectation that lower similarity (SCI) corresponds to higher discrepancies (EIM). For example, Tree SHAP starts with near-zero MAE for all classes at a 0.5% noise level and maintains MAEs of around 0.02% for all classes at perturbation levels up to 5%. Kernel SHAP shows higher MAEs, with

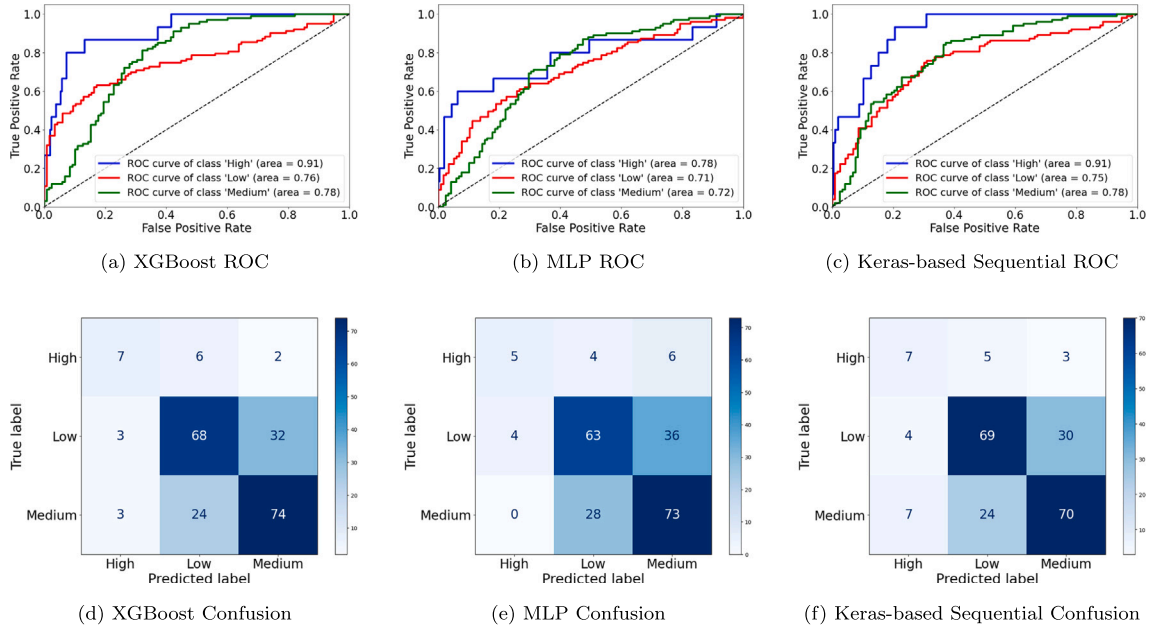


Fig. 3. Comparison of ROC curves (top row) and confusion matrices (bottom row) for XGBoost, MLP, and Keras-based Sequential models.

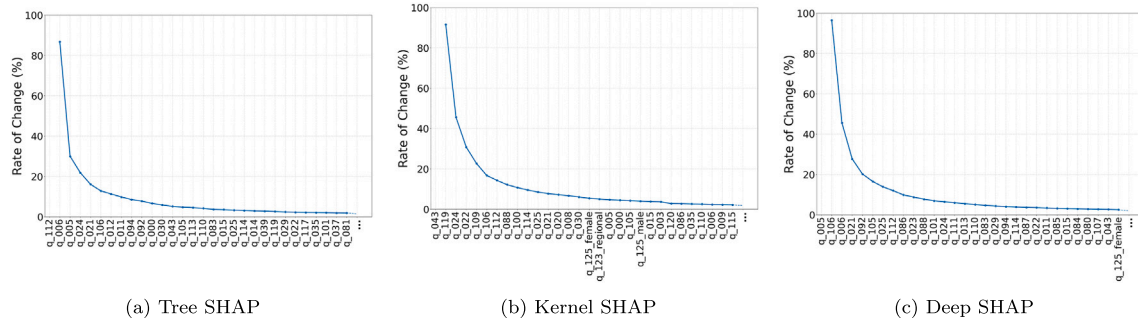


Fig. 4. Determining important features based on rates of change in cumulative contributions. Note that we only displayed the top 30 features here. The full set is available in Fig. D1.

approximately 0.22% for the Low class, 0.58% for the Medium class, and 0.56% for the High class at a 5% perturbation level. In contrast, the Deep Explainer exhibits relatively high EIM scores across all noise levels, though they remain relatively constant as perturbation levels increase beyond 5%.

Therefore, we conclude that the Tree Explainer provides the most stable explanations upon noises, which is closely followed by the Kernel Explainer. The Deep Explainer seems vulnerable to noise, as the explainer relies on output gradients with respect to the input features (Shrikumar et al., 2016), which can vary significantly with small perturbations. This comparative analysis highlights the critical need for selecting appropriate explainers based on the model architecture and the specific requirements of the explanatory task at hand.

4.3. Factors influencing the level of participation in CRE

Fig. 7 presents SHAP-based visualizations—including summary plots, decision plots, and a global bar plot—generated using the TreeExplainer for low, medium, and high CRE participation levels. These visualizations illustrate how different features contribute to model predictions across and within participation classes. To enhance the interpretability of these patterns, Appendix D.2 identifies features that consistently influence predictions across multiple classes, while Appendix D.3 presents

force plots to provide localized, respondent-specific explanations from the XGBoost-based TreeExplainer.

The global summary bar plot (Fig. 7(a)) provides a cumulative view of feature influence, revealing which variables consistently contribute to model predictions across all classes. For instance, policy risk (q_{112}), perception of corporate RE and CRE effectiveness (q_{006}), effectiveness of community-owned projects (q_{005}), and intentions to install RE (q_{024}) are among the most influential, indicating their widespread importance in shaping participation regardless of level.

A comparison across the summary plots (Fig. 7(b), (d), (f)) reveals that several features exhibit bipolar influence—appearing prominently in both Low and High participation classes but not in Medium. This pattern suggests that certain variables may act as threshold or tipping point factors, polarizing individuals into either disengagement or full participation depending on their feature values.

For example, plans to install renewable technologies (q_{024}) emerge as a critical factor, with a clear directional impact. SHAP values range from approximately -0.3 to 0 for low participation and 0 to 0.2 for high participation. This indicates that as the intention to adopt RE increases, participation shifts strongly from low to high. This factor is notably absent in medium participation, suggesting a polarizing effect that pushes individuals toward either end of the participation spectrum. The effectiveness of small/medium community-owned projects (q_{005}) shows a

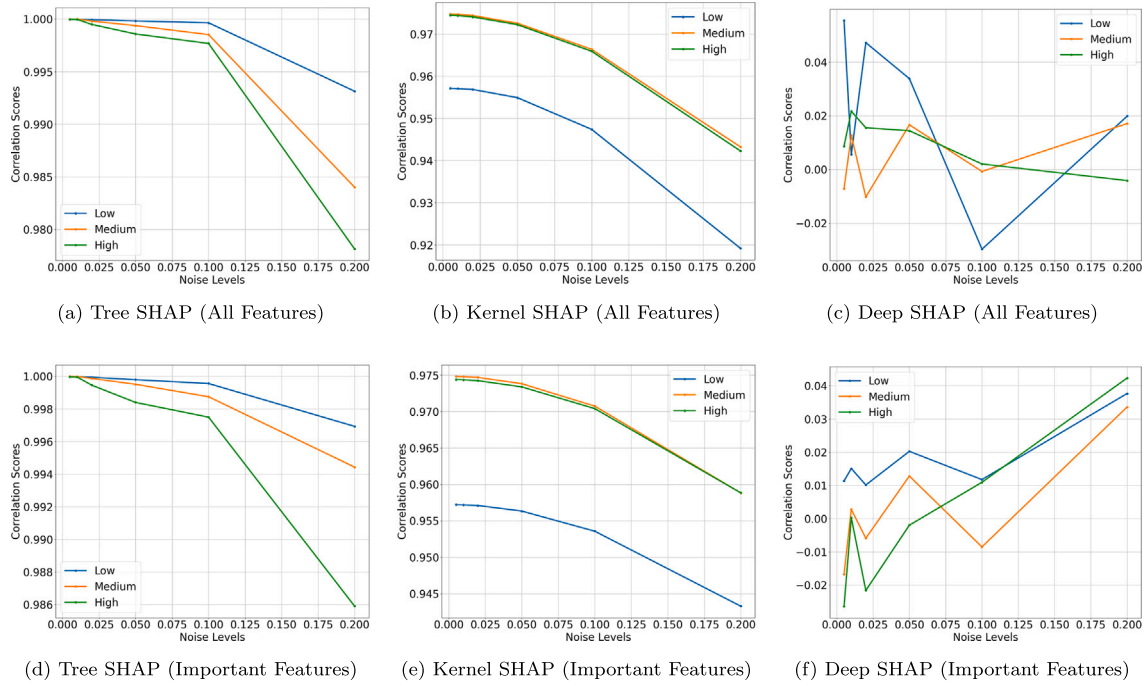


Fig. 5. Comparison of Tree SHAP, Kernel SHAP, and Deep SHAP based on SCI correlation. The top row shows performance over the entire feature space; the bottom row focuses on selected important features.

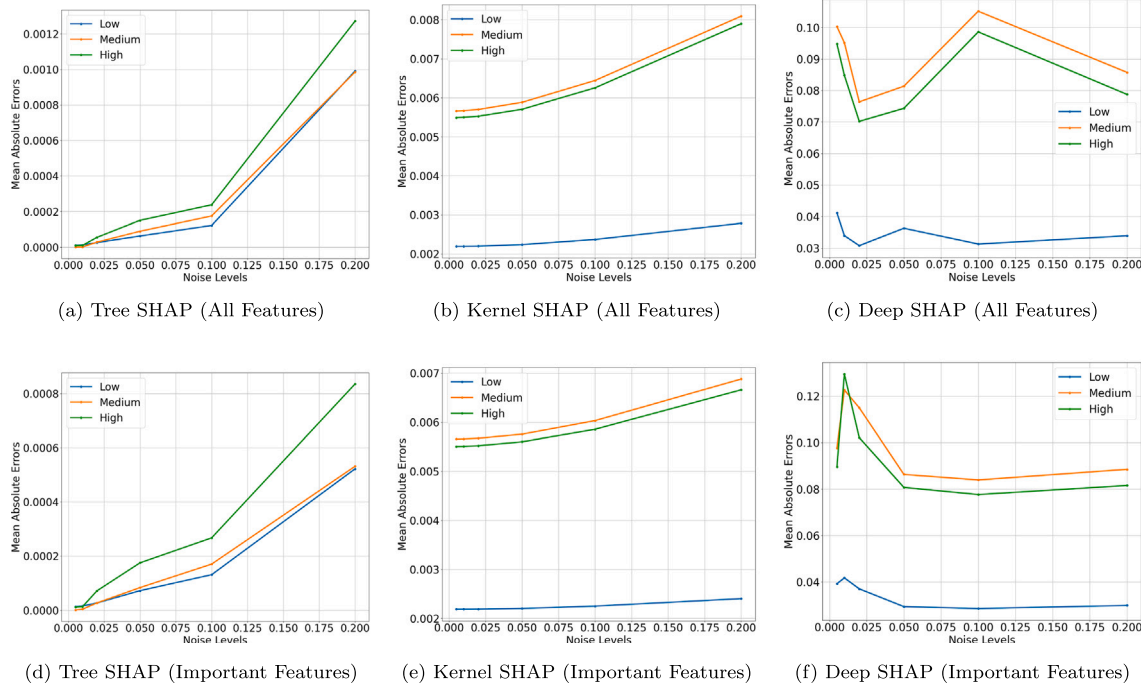


Fig. 6. Comparison of tree SHAP, Kernel SHAP, and deep SHAP based on EIM (measured by MAE). The top row evaluates the entire feature space, while the bottom row focuses on important features only.

similar pattern, with SHAP values of about -0.2 to 0 for low participation and 0 to 0.3 for high participation, suggesting that greater belief in the efficacy of local initiatives significantly boosts engagement levels from low to high.

Similarly, local participation in energy decisions (q_{106}) and existing RE installations (q_{021}) both exhibit a positive directional impact. Lower values contribute to low participation, while higher values promote high

participation. This suggests a linear relationship between these factors and participation levels. Overcoming barriers to RE production (q_{025}) shows a complex impact, with SHAP values ranging from approximately -0.2 to 0 for low participation and 0 to 0.1 for high participation. This indicates that as perceived ability to overcome barriers increases, participation tends to shift from low to high, albeit with a modest effect at the high end.

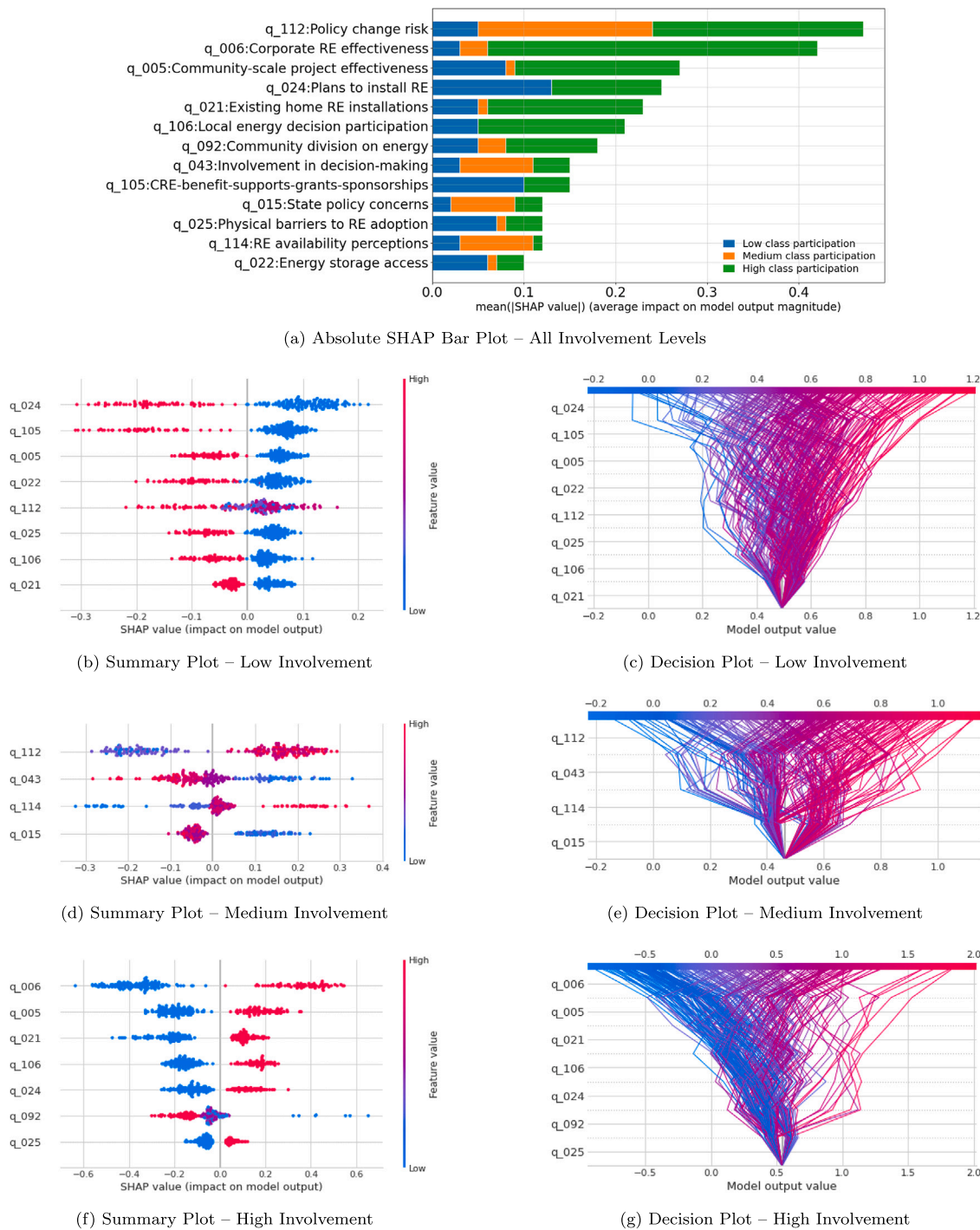


Fig. 7. Multiple SHAP visualizations grouped by participation levels. **Top:** a global bar plot showing which features are generally most influential in the model predictions, regardless of class. **Left:** summary plots show how each feature affects predictions for a specific class (Low, Medium, or High). Red means a high value for that feature, blue means a low value. Dots on the right mean the feature made the model more confident in predicting that class, while dots on the left mean it pushed the model away from that class. **Right:** Decision plots visualize how features combine to make a prediction for a specific class. Each line is a respondent. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Some factors exhibit more complex directional effects. The risk of government policy changes (q_{112}) shows mixed impacts, with SHAP values ranging from approximately -0.1 to 0.1 for low participation and -0.3 to 0.3 for medium participation. Higher values appear to move individuals from low to medium participation, indicating that increased

policy risk awareness may motivate moderate engagement. This suggests that increasing awareness of the risk of changing policy pushes individuals from low to medium participation. The importance of being involved in decision-making (q_{043}) shows an interesting directional impact: lower values promote medium participation, while higher values

likely push toward high participation (though not directly shown in the data). This suggests that as the perceived importance of involvement increases, participation levels tend to rise. Beliefs about RE availability (q_{114}) have mixed directional impacts on medium participation, with SHAP values spanning from roughly -0.3 to 0.3 , indicating that this factor's influence depends heavily on individual or local contexts. Concerns about state government policies (q_{015}) primarily impact medium participation, with SHAP values mostly in the positive range (0 to 0.2), indicating that higher concerns about state policies tend to promote medium levels of engagement.

Certain factors appear to have specific impacts on particular participation levels. Community sponsorship/grants (q_{105}) primarily influence low participation, with SHAP values from about -0.3 to 0 , suggesting that limited access to community funding significantly reduces engagement. Energy storage solutions (q_{022}) show a similar pattern of low participation, with SHAP values of approximately -0.3 to 0 , indicating that limited knowledge or resources in this area hinder engagement. Conversely, the effectiveness of medium/large corporate-owned projects (q_{006}) emerges as a dominant factor specifically for high participation, with SHAP values ranging from approximately 0.2 to 0.6 , although its influence is asymmetric. Some factors, such as community divisions about energy sources (q_{092}), show minimal impact on participation levels, with SHAP values clustered around zero (-0.1 to 0.1) for high participation. This suggests that certain community-level factors may not be as influential as individual perceptions or tangible project-related factors in determining engagement levels.

In the decision plots (Fig. 7(c), (e), (g)), individual lines represent respondents. Lines that rise sharply indicate strong cumulative support toward class membership, while downward trends show suppressing effects. Features are ordered top-to-bottom by average position in the prediction trajectory. Some features appear to act as entry-point bottlenecks—most notably, access to community funding (q_{105}) and energy storage (q_{022}), which show stronger influence on low participation but minimal impact elsewhere. Other features, such as existing RE use (q_{021}) and policy concerns (q_{112}), span across classes, indicating overlapping drivers. These patterns suggest the importance of distinguishing between foundational enablers for initial involvement and broader factors that sustain or deepen engagement.

5. Discussion

This study makes theoretical and practical contributions by precisely mapping the spectrum of roles and engagement levels within CRE projects, providing actionable insights to enhance involvement.

5.1. Methodological implications for XAI

From a methodological perspective, our study revealed that XGBoost and other tree-based models outperformed neural network approaches like MLP in predicting CRE project participation levels. This suggests that the underlying factors influencing participation may have a hierarchical or non-linear structure that tree-based models capture more effectively. Interestingly, all models demonstrated the highest ROC-AUC values for the minority 'High' participation class (91% for XGBoost and Keras-based Sequential, 78% for MLP). This unexpected finding indicates that, despite a lower overall number of true high-participation instances, the models are particularly effective at distinguishing high-participation cases across different decision thresholds. While models made more true positive predictions for the 'Medium' and 'Low' classes, these predictions appeared to involve more 'guessing' rather than a clear distinction, as evidenced by lower ROC-AUC values. This highlights a potential misalignment between prediction frequency and confidence for these classes, emphasizing the importance of considering multiple performance metrics in imbalanced multi-class scenarios.

The developed metrics (EIM and SCI) revealed significant differences in the robustness of the SHAP explainers. Tree SHAP and Kernel SHAP maintained high correlation coefficients (above 90%) even under increasing noise levels, while Deep SHAP showed surprising vulnerability to even minimal perturbations of 0.5%, resulting in correlation drops to near-zero or negative values. This finding challenges the assumption that all SHAP methods are equally reliable in different model architectures. This aligns with observations by Ancona et al. (2019) regarding instability issues typical of gradient-based explanations, such as high sensitivity to input changes and local gradient information (Ancona et al., 2019). Our findings raise questions about the reliability of gradient-based explanation methods in complex, non-linear models when applied to noisy real-world data. Additionally, the analysis revealed that focusing on a subset of important features (selected through a 3% rate of change threshold) not only reduced computational complexity but actually improved the stability of explanations, as evidenced by higher SCI scores. This challenges the common assumption that more features necessarily lead to more robust explanations and suggests that feature selection based on contribution patterns might enhance both efficiency and reliability.

Beyond explainer robustness, our analysis revealed asymmetric feature importance across participation levels. Government policy risk (q_{112}) emerged as a critical predictor for medium participation but showed minimal influence on high engagement, while corporate project effectiveness (q_{006}) strongly drove high participation yet barely registered for low. These asymmetries were detectable because Tree SHAP leverages model structure directly, avoiding the gradient instability that causes Deep SHAP to collapse under minimal perturbation. Furthermore, our selective perturbation approach—targeting influential features identified by SHAP rather than applying uniform noise—enabled meaningful robustness testing by focusing on what actually matters for prediction. This filtering of signal from noise proved particularly valuable for heterogeneous behavioral data. Class-specific SHAP decomposition revealed patterns that global importance rankings would otherwise average away. Traditional regression coefficients assume uniform effects across outcomes, and permutation importance treats all features equally regardless of their differential roles across classes. Our approach succeeds by accounting for the structure of both the model and the data. Tree architectures capture nonlinear behavioral patterns, targeted perturbations provide meaningful robustness validation, and class-level analysis reveals asymmetric determinants that inform stage-specific intervention design.

5.2. Theoretical implications for participation in CRE

From a theoretical perspective, our analysis demonstrates that different sets of factors are crucial in predicting low, medium, and high participation as shown in Fig. 7. One of the most surprising findings is the polarizing impact of plans to install renewable technologies (q_{024}). Individuals who have strong plans to adopt renewable technologies exhibit higher participation levels, while those without such plans tend to remain disengaged. This polarized outcome suggests that, contrary to expectations, people are either all-in or not involved at all, with little room for moderate engagement. This finding challenges the assumption that gradually increasing awareness will lead to medium-level participation (Chulow et al., 2021). Instead, it indicates that people either see significant personal value in these technologies and engage deeply or do not engage at all, making it difficult to target those in the middle.

Interestingly, the acknowledgment of governmental policy change risks (q_{112}) did not uniformly deter participation as one might expect. Instead, it appeared to indicate a more informed engagement with the energy sector, particularly among those showing medium participation levels. This finding challenges the conventional wisdom that perceived policy instability invariably leads to lower participation (Polzin et al.,

2019; Brauholtz-Speight et al., 2020). It suggests that a certain level of risk awareness might actually be associated with a more committed and knowledgeable form of engagement, possibly due to a desire to hedge against these risks or influence policy changes.

The observed role of government policy risk as both a motivator and a deterrent highlights the fragile interplay between regulatory environments and citizen trust (Taghikhah et al., 2020, 2019). This implies that policy stability and transparency are not merely administrative goals but essential conditions for fostering community engagement. Thus, policymakers should implement participatory frameworks, embedding citizen voices at the heart of energy governance (Hafezi et al., 2023). This could include town hall meetings, public consultations, or even local energy councils that work directly with governments on policy formation.

Another controversial result comes from the perception of the effectiveness of corporate-owned projects. Typically, large-scale energy initiatives are assumed to draw public skepticism in favor of smaller community projects. However, Shapley values for q_{006} indicate that a strong belief in the effectiveness of corporate-owned projects can push people to either high or low participation extremes. Far from being merely a source of skepticism, such projects represent a latent opportunity for hybrid models that balance the operational efficiency of large-scale enterprises with the trust and inclusivity of community-led initiatives. This insight shifts the narrative, showing that corporations, often viewed as external actors (Cohen et al., 2021), can become allies in fostering local energy independence if appropriately aligned with community values and governance structures.

Lastly, the limited impact of community divisions about energy sources (q_{092}) on participation levels was unexpected. It was assumed that local disagreements over energy sources would significantly hinder engagement (Mey et al., 2016; Shobande et al., 2023). However, the SHAP values indicate minimal influence, particularly for high participants. This suggests that while community divisions exist, they may not be as impactful on actual participation levels as previously thought (Sattayapanich et al., 2022), indicating that individual motivations and external factors like funding and project efficiency may outweigh local conflicts in determining involvement. Policymakers may shift attention away from local divisions and emphasize the tangible benefits of participation in CRE projects, such as cost savings, energy security, and environmental benefits (Hoicka and MacArthur, 2018).

5.3. Policy implications

Australia's CRE sector operates within a fragmented regulatory landscape where federal frameworks such as the Renewable Energy Target intersect with variable state-level feed-in tariffs and grid connection requirements (Byrnes et al., 2013). Community energy projects face additional hurdles including electricity retail licensing, inconsistent planning approvals across jurisdictions, and policy instability—exemplified by feed-in tariff reductions in Queensland (2012) and South Australia (2013) that disrupted projects mid-development (Simpson and Clifton, 2017). Organizations such as the Community Power Agency and Coalition for Community Energy have advocated for regulatory reform, including dedicated community energy targets and streamlined licensing (Community Power Agency, 2021). Our findings should be interpreted within this context, and the policy recommendations below are designed to address barriers specific to the Australian regulatory environment.

Table 3 presents four integrated policy bundles derived from SHAP-based analysis of the key drivers of CRE participation, with explicit model evidence, recommended interventions, and Australian regulatory context for each bundle. The identified groupings—*Home RE Support*, *Community Engagement and Governance*, *Institutional Trust and Stability*, and *Social Norms and Peer Influence*—operate across multiple scales, ranging from individual household choices to institutional frameworks and community-level social dynamics. Additional policy

recommendations mapped to individual features and participation levels are provided in Appendix E.

Each policy bundle represents a different pathway to participation—material, social, institutional, or normative—and our SHAP analysis reveals distinct patterns for each. The *Home RE Support* bundle targets material constraints, with physical adoption barriers (q_{025}) consistently showing negative effects (SHAP -0.10 to -0.20) across all participation levels, while RE installation plans (q_{024}) exhibit polarizing effects (SHAP -0.15 to $+0.35$), facilitating engagement when infrastructure exists but impeding it when barriers persist. These findings suggest that addressing material constraints offers a fast pathway to shift low-participation groups. The *Community Engagement and Governance* bundle addresses political and relational dimensions. Community sponsorship opportunities (q_{105}) show an asymmetric pattern, with positive SHAP values for low participation but near-zero for high, indicating that grant-based framing attracts initial interest but does not sustain advanced engagement. This bundle demands longer time horizons but is essential for embedding durable community ownership.

The *Institutional Trust and Stability* bundle reveals our most striking asymmetric finding, with policy risk perception (q_{112}) strongly predicting medium participation (SHAP $+0.15$ to $+0.25$) but showing minimal effect on high participation. This suggests that regulatory uncertainty disproportionately affects citizens on the threshold of deeper engagement—precisely those whom policy should target. Corporate RE effectiveness (q_{006}) drives high participation exclusively (SHAP $+0.20$ to $+0.60$), indicating that confidence in large-scale projects complements rather than substitutes for community engagement. The *Social Norms and Peer Influence* bundle leverages normative factors. Peer RE adoption (q_{094}) shows consistent positive effects across all participation levels (SHAP $+0.05$ to $+0.12$), confirming social proof mechanisms documented in prior research. Community energy divisions (q_{092}) exhibit a polarizing pattern—predicting both very low participation (where conflict suppresses engagement) and very high participation (where conflict motivates activism)—suggesting context-specific interventions are required.

Together, these bundles provide a roadmap for layered and synchronized action. Accelerating participation in CRE requires more than offering technology or financial incentives. It demands the careful orchestration of material support, institutional credibility, community empowerment, and socially visible engagement. The asymmetric effects we document challenge universal policy approaches, indicating that effective interventions must be stage-specific, addressing material barriers for low-engagement citizens, building institutional trust for medium-engagement citizens, and providing governance pathways for high-engagement citizens.

These findings point to a fundamental rethinking of national energy policies, challenging the conventional reliance on universal solutions and homogeneous approaches. Instead, they highlight the imperative for adaptable systems that honor the unique contexts of communities while connecting them to broader energy infrastructures. This perspective is especially critical for regions marked by profound social and economic disparities, where failing to address local realities risks deepening inequalities under the pretense of sustainability. Effective energy systems of the future must balance scalability with inclusivity, demonstrating that solutions capable of broad application also deeply engage those they aim to serve. This duality—scalable yet participatory—is not just an ideal but a foundational principle for tackling the complex, interconnected challenges of climate change, energy equity, and policy responsiveness. The study ultimately urges stakeholders—policymakers, corporations, and communities alike—to reframe energy transitions as deeply human processes rather than merely technical or economic pursuits. True long-term energy equity will emerge only by embedding participation as an intrinsic element of energy systems, making it a cornerstone of design and implementation rather than an afterthought.

Table 3
Integrated policy packages: Model evidence, recommendations, and Australian context.

Policy bundle	Model evidence (SHAP analysis)	Recommended policy	Australian regulatory context	Target shift	Time
Home RE Support <i>Material focus</i>	• q.024 (RE installation plans): SHAP −0.15 to +0.35; polarizing effect • q.025 (physical barriers): SHAP −0.10 to −0.20; consistent negative across all levels • q.022 (battery storage): SHAP +0.12 to +0.28 for high participation	Zero-interest solar/battery financing; targeted storage subsidies; renter-specific solutions (solar leasing, shared rooftop); RE Ambassador programs for peer guidance	Existing programs: Victoria Solar Homes Program (interest-free loans for solar/batteries); SA Home Battery Scheme (subsidies up to \$6,000); ACT Sustainable Household Scheme (Australian Energy Regulator, 2023) Policy gap: Renters (31% of Australian households) lack adoption pathways (Australian Bureau of Statistics, 2021); strata/apartment schemes remain underdeveloped	Low → High	Medium
Community Engagement & Governance <i>Social focus</i>	• q.005 (community project effectiveness): SHAP +0.10 to +0.25 for high participation • q.106 (local decision participation): SHAP +0.08 to +0.18 • q.105 (sponsorship/grants): SHAP +0.15 for low, near-zero for high (<i>asymmetric</i>)	Local energy committees with decision-making authority; startup grants; open planning workshops; leadership pathways for engaged citizens; evolve messaging from “funding” to “ownership”	Existing models: Hepburn Wind (Australia's first community-owned wind farm, VIC); Enova Energy (community retailer, NSW) (Gui and MacGill, 2017); CEFC community grants Policy gap: No national community energy target; electricity retail licensing barriers; inconsistent state-level support (Community Power Agency, 2021)	Low/Med → High	Long
Institutional Trust & Stability <i>Institutional focus</i>	• q.112 (policy risk): SHAP +0.15 to +0.25 for medium only (<i>key asymmetric finding</i>) • q.015 (state policy concerns): SHAP +0.08 to +0.15 for medium • q.006 (corporate RE effectiveness): SHAP +0.20 to +0.60 for high only	Bipartisan long-term policy roadmaps; incentive stability with grandfathering provisions; public education on 24/7 RE technologies; corporate RE benefit communication (jobs, grid reliability)	Historical context: RET reduction (2015) created investment uncertainty; QLD and SA FiT cuts (2012–13) disrupted community projects mid-development (Byrnes et al., 2013 ; Simpson and Clifton, 2017) Policy need: Regulatory certainty mechanisms; longer review cycles; dedicated community energy carve-outs within federal frameworks	Low/Med → Med/High	Long
Social Norms & Peer Influence <i>Normative focus</i>	• q.094 (peer RE adoption): SHAP +0.05 to +0.12; consistent across all levels • q.092 (community divisions): high values predict both very low and very high participation (<i>polarizing</i>) • q.121 (usage change from savings): SHAP +0.06 to +0.10	Peer adoption storytelling; neighborhood RE champions; energy “open house” events; community deliberation forums; peer mentor mobilization	Existing initiatives: Australian Energy Foundation community programs; Neighbourhood Houses Victoria; local council sustainability programs (Australian Energy Foundation, 2022) Opportunity: Coordinated “solar streets” campaigns leveraging visible adoption narratives	Low → Medium	Medium

Notes: Abbreviations: SA = South Australia State; ACT = Australian Capital Territory; QLD = Queensland State; VIC = Victorian State; NSW = New South Wales State; RET = Renewable Energy Target; FiT = Feed-in Tariff; CEFC = Clean Energy Finance Corporation; CPA = Community Power Agency; C4CE = Coalition for Community Energy.

6. Conclusions

This study addressed three research questions concerning citizen participation in CRE initiatives. Regarding RQ1—whether ML models can accurately classify citizens into distinct participation levels—XGBoost achieved 68% accuracy and 81% macro ROC-AUC, with particularly strong performance in distinguishing high participation cases (91% ROC-AUC). This demonstrates that ML approaches can effectively capture the nonlinear, multidimensional nature of CRE engagement that traditional statistical methods struggle to model. Regarding RQ2—the robustness of SHAP-based explanations for heterogeneous behavioral data—our novel SCI and EIM metrics revealed that Tree SHAP maintained stability above 99% under moderate perturbation, while Deep SHAP collapsed under minimal noise. This finding challenges the assumption that all SHAP methods are equally reliable and underscores the importance of robustness validation before interpreting behavioral data. Regarding RQ3—factors driving varying engagement levels—SHAP analysis revealed asymmetric determinants: plans to adopt renewable technologies polarize participation into high or low extremes, corporate project perceptions selectively drive high engagement, and policy

risk awareness motivates moderate participation rather than deterring it. The SCI and EIM metrics, while developed for CRE participation, hold potential for broader application in domains where human motivation is central but difficult to model—such as climate adaptation, public health, or civic technology.

Some considerations should be noted. Firstly, while the sample size is substantial, it is skewed toward younger respondents, which may bias findings toward the attitudes and behaviors more common in this demographic—such as higher digital engagement or climate awareness. This age skew, however, may also reflect the reality that younger people are more engaged with RE and environmental issues, as seen in prior work highlighting youth as key catalysts for climate action ([Pandve et al., 2009](#); [Mission Australia, 2023](#); [World Economic Forum, 2023](#)). As a result, our findings may underrepresent the perspectives of older adults, who may face distinct barriers or motivations. Additionally, remote and Indigenous communities were underrepresented, which limits the generalizability of the results across all Australian regions and social contexts. As with most social science survey research, self-report bias is a potential concern, though our use of actual behavioral roles rather

than intentions may mitigate some response biases. The survey, while comprehensive, did not capture specific data on the amount of financial investment or time respondents contributed to CRE projects. Moreover, our focus on individual-level variables may overlook community-level dynamics or structural conditions that shape participation. Although the 3% feature threshold used in our analysis was informed by empirical patterns, it may not generalize across datasets and warrants further methodological validation. Nonetheless, our study is among the first to combine multi-class ML classification with SHAP-based robustness analysis to understand behavioral drivers of CRE participation. This methodological integration opens new avenues for research into interpretable, equitable, and data-driven energy policy design.

Future research may employ longitudinal studies to track changes in attitudes and participation over time. Expanding the survey to include more diverse communities and gathering more granular data on financial capabilities and local energy infrastructure would provide a more comprehensive picture. Improving demographic representation through stratified sampling would also help correct potential age-related biases in the dataset. Triangulating self-reported measures with CRE project membership records or administrative data would strengthen construct validity and enable assessment of potential response biases. In addition, investigating potential moderating impacts of demographics or social roles using causal inference methods could offer deeper insights into group-specific participation drivers. The assignment of weights to different roles can be adjusted based on their complexity, time commitment, impact on project outcomes, or perceived importance of each role to community members. Additionally, incorporating qualitative methods such as in-depth interviews or focus groups could offer deeper insights into the decision-making processes that our quantitative approach may have missed. Future studies could also benefit from linking survey responses to actual participation data in CRE projects, rather than relying solely on self-reported intentions or past behaviors.

CRedit authorship contribution statement

Firouzeh Rosa Taghikhah: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Arunima Malik:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Funding acquisition, Conceptualization. **Alexey Voinov:** Writing – review & editing, Writing – original draft, Validation, Data curation, Conceptualization. **Kannan Govindan:** Writing – review & editing, Writing – original draft, Validation. **Babak Mohamadpour Tosarkani:** Writing – review & editing, Visualization, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Survey design and distribution

The survey was meticulously designed to capture a comprehensive understanding of the factors influencing participation levels in CRE initiatives. It integrates key elements from Social Identity Theory (Tajfel, 1979), the Theory of Planned Behavior (Ajzen, 1991), and the Norm Activation Model (Schwartz, 1977). In total, the survey includes 11 constructs measured through 90 items.

Social Identity Theory is applied through questions that gauge respondents' sense of community allegiance and attitudes toward CRE projects, reflecting the theory's emphasis on group membership as a source of identity and motivation. For instance, items assessing connection to the community (e.g., "I feel strongly connected to the community I live in") and attitudes toward community energy ownership (e.g., "I like the idea of a community owning and being in control of its energy supply") tap into the social identity aspects.

The Theory of Planned Behavior is incorporated through items that assess attitudes toward RE, subjective norms, and perceived behavioral control. Attitudinal measures include statements about the sensibility and emotional appeal of RE (e.g., "I think using renewable energy is a sensible idea", "Renewable energy gives me a good feeling"), which are part of the construct *Attitude to Renewable Energy*.

Subjective norms are addressed through questions about peers' adoption of RE technologies, such as "Have any of your friends, neighbours or others you know already got household renewable energy?", included in the construct *Social Cultural Norms*. Perceived behavioral control is reflected in items that explore barriers to participation in CRE projects (e.g., "The financial costs of joining a community energy project are too high", "Joining a community energy project may demand too much effort from individuals") and willingness to invest in RE technologies (e.g., "I plan to install renewable within the next 1–2 years", "I want to generate more renewable energy but currently have barriers").

The Norm Activation Model focuses on personal norms, awareness of consequences, and ascription of responsibility. It is represented through questions that assess personal feelings of responsibility for climate change (e.g., "Personally, I feel responsible for climate change"), awareness of environmental problems (e.g., "Environmental problems are frequently exaggerated"), and perceived benefits of RE adoption (e.g., "Acting on climate change/carbon emissions reductions", "Reduce the risk of destruction of land and water"). The survey also includes items that gauge respondents' perceptions of the benefits of switching to RE, reflecting personal ascription of responsibility for energy transition.

For the dependent variable, respondents were asked to indicate their participation across multiple roles: Investor (*q_045*) contributing financially to the project; Employed or Contracted (*q_046*) working in a paid or contracted capacity; Volunteer (*q_047*) assisting voluntarily, from key organizing to event setup; Decision-Making Group (*q_048*) participating in boards or committees that make project decisions; Occasional Expertise (*q_049*) providing expert advice or serving on advisory groups sporadically; Traditional Custodian Advisor (*q_050*) offering insights based on traditional and local knowledge; Community Engagement (*q_051*, *q_052*) communicating about the project with locals or discussing it with government and business stakeholders; Resource Provision (*q_053*) providing or leasing land or spaces for the project; Supply of Inputs (*q_054*) contributing additional resources like crop waste for bioenergy.

The survey was distributed through Geni.Energy and Community Power Agency networks in 2023, yielding 1014 initial responses from Australian residents. To ensure data quality, we implemented several screening criteria: (1) completion time filters excluded surveys completed in less than 8 minutes (indicating insufficient engagement) or

Table A1
Demographic information of survey respondents.

Variable	Category	Frequency (N)	Percentage (%)
Gender	Male	473	54%
	Female	402	46%
Age	18 to 34 years	499	57%
	35 to 54 years	298	34%
	55 to 74 years	61	7%
	75 years and over	17	2%
Income	Less than \$30,000	200	23%
	\$30,000 to \$60,000	280	32%
	\$60,000 to \$100,000	260	30%
	\$100,000 to \$150,000	100	11%
	More than \$150,000	35	4%
Living Area	Capital city areas	464	53%
	Rest of state/territory areas	411	47%
Living State	New South Wales (NSW)	289	33%
	Victoria (VIC)	184	21%
	Queensland (QLD)	193	22%
	Western Australia (WA)	70	8%
	South Australia (SA)	70	8%
	Tasmania (TAS)	35	4%
	Australian Capital Territory (ACT)	8	1%
	Northern Territory (NT)	26	3%

Table A2
Factor loadings summary.

Construct	Items	Loading range
Attitude to CRE	q.080–q.085	0.58–0.84
Environmental Belief	q.086–q.088	0.61–0.82
Perceived Barriers to CRE	q.110–q.112, q.114–q.120	0.46–0.79
Perceived Barriers to RE	q.011–q.018	0.49–0.77
Willingness to Invest in RE	q.020–q.025, q.028, q.029, q.036, q.031, q.034, q.035	0.44–0.86
Community Allegiance	q.089–q.093	0.57–0.83
Social Cultural Norms	q.094, q.113, q.121	0.66–0.88
Perceived Benefits of CRE	q.037–q.044, q.099–q.109	0.45–0.81
Achievement of RE Target	q.000–q.001, q.003–q.009	0.44–0.80
Roles in CRE Projects	q.045–q.054	0.47–0.83
Demographics	q.123, q.125, q.126, q.130	n/a

more than 60 minutes (indicating interruptions or inattention), resulting in 47 exclusions; (2) two instructional manipulation checks were embedded (e.g., “Please select ‘Strongly Agree’ for this item”), with respondents failing both checks excluded ($N = 38$); (3) surveys with incomplete responses on key variables were excluded ($N = 54$). After applying these criteria, the final analytical sample comprised 875 respondents. Median completion time for the retained sample was 18 minutes (range: 12–47 minutes).

Demographic information from survey respondents is given in Table A1. The table summarizes the key socio-demographic variables collected in the survey, including gender, age, living area, and living state.

As shown in Table A1, the survey sample contains a disproportionately higher number of younger respondents. This skew may reflect

increased digital accessibility and stronger climate engagement among youth—groups widely recognized as catalysts for environmental action. While this enhances the model ability to capture youth-driven participation patterns, it may underrepresent the perspectives of older populations who might face distinct motivations or barriers to CRE engagement. To mitigate this potential bias, our machine learning models capture complex, non-linear relationships across features without assuming homogeneous behavior across age groups. Furthermore, SHAP-based interpretability techniques allow us to quantify the influence of demographic variables such as age, offering transparency into how such skews affect prediction outcomes. These insights can guide future work in implementing demographic weighting, stratified sampling, or targeted outreach to achieve more balanced representation.

The reliability of the survey instrument was assessed using Cronbach’s alpha coefficient (Nunnally and Bernstein, 1994), which demonstrated high internal consistency across all constructs: Attitude to RE ($\alpha = 0.912$), Attitude to CRE ($\alpha = 0.905$), Perceived Barriers to RE ($\alpha = 0.850$), Perceived Barriers to CRE ($\alpha = 0.890$), Perceived Benefits of CRE ($\alpha = 0.900$), Social Cultural Norms ($\alpha = 0.860$), Community Allegiance ($\alpha = 0.870$), Willingness to Invest in RE ($\alpha = 0.895$), and Roles in CRE ($\alpha = 0.860$). These high alpha values (all above 0.8) indicate excellent reliability of the survey instrument, suggesting that the items within each construct effectively measure the intended concepts and providing a solid foundation for further analysis of CRE participation factors.

Exploratory Factor Analysis was conducted to verify that items loaded appropriately onto their theoretical constructs. The Kaiser–Meyer–Olkin measure of sampling adequacy was 0.89 and Bartlett’s test of sphericity was significant ($\chi^2 = 11,420.3$, $p < 0.001$), confirming data suitability for factor analysis. Principal axis factoring with Promax rotation revealed a 10-factor structure (excluding demographics) consistent with the theoretically derived constructs. Items with factor loadings below 0.40 or cross-loadings exceeding 0.30 were reviewed; all retained items met these thresholds. Table A2 presents the factor loading matrix.

Appendix B. Descriptive statistics of survey variables

We calculated skewness, kurtosis, and Kolmogorov–Smirnov test statistics for each survey variable to assess conformity to normal distribution assumptions. Table B1 reports the distributional characteristics of all variables, including p-values indicating deviations from normality. As expected in attitudinal survey data, many variables exhibit non-normal distributions, with mild to moderate skewness and mesokurtic tendencies. Table B2 summarizes key descriptive statistics for the subset of features used in ML.

While many variables deviate from normality, we intentionally employed ML algorithms—such as XGBoost, MLP, and Keras Neural Networks—that are robust to such deviations. These models do not rely on assumptions of linearity, normality, or homoscedasticity and are capable of capturing complex, non-parametric relationships in behavioral data. Thus, distributional skew and kurtosis were addressed through algorithm selection rather than data transformation. To ensure model reliability and reduce overfitting, we applied stratified sampling, class weighting, and hyperparameter optimization across all models. We recognize, however, that some variables—such as perceptions of institutional responsibility or personal climate agency—may reflect underlying social or cultural biases. Although we did not correct for these explicitly, their influence is explored through post hoc model interpretation using SHAP, which reveals their relative contribution to predictions across participation levels.

Table B1

Survey description: skewness, kurtosis, and normality tests.

No.	Question	Label	Variable-items	Skewness	Kurtosis	Kolmogorov-Smirnov p-value
Attitude to CRE						
1	q_083	CRE-attitude-environmental-exaggeration	Environmental problems are frequently exaggerated	−0.50	−0.45	0.00
2	q_081	CRE-attitude-sensible-choice	I think using renewable energy is a sensible idea	−0.46	−0.43	0.00
3	q_082	CRE-attitude-positive-emotion	Renewable energy gives me a good feeling	−0.39	−0.27	0.00
4	q_084	CRE-attitude-informed	I consider myself well-informed about renewables and energy	−0.30	−0.37	0.00
5	q_080	CRE-attitude-ownership	I like the idea of a community owning and controlling its energy supply	−0.48	−0.11	0.00
6	q_085	CRE-attitude-community-overlooked	My community is overlooked by policymakers when upgrading grid infrastructure	−0.30	−0.45	0.00
Environmental belief						
7	q_086	climate-belief-personal-responsibility	Personally, I feel responsible for climate change	−0.66	0.25	0.00
8	q_087	climate-belief-energy-use-irrelevant	The way I personally use energy does not make a difference to pollution	−0.58	−0.29	0.00
9	q_088	climate-belief-trust-government	I trust the government to take positive action on pollution and climate change	−0.69	0.20	0.00
Perceived Barriers to CRE						
10	q_110	CRE-barrier-costs-high	The financial costs of joining a community energy project are too high	−0.45	0.02	0.00
11	q_111	CRE-barrier-effort-high	Joining a community energy project may demand too much effort	−0.32	−0.20	0.00
12	q_112	CRE-barrier-policy-risk	Joining a community energy project is risky because the government may change the rules	−0.38	−0.19	0.00
13	q_114	CRE-barrier-unavailable-night	Renewable energy is not available at night	−0.09	−0.94	0.00
14	q_115	CRE-barrier-jobs-not-local	Jobs in renewables are not local	0.12	−0.92	0.00
15	q_116	CRE-barrier-profits-external-investors	Profits from renewables go to external investors	0.02	−0.75	0.00
16	q_117	CRE-barrier-supports-regional-development	Renewables will help regional development	−0.24	−0.92	0.00
17	q_118	CRE-barrier-hinders-farming	Large-scale renewable projects may hinder farming	0.01	−0.92	0.00
18	q_119	CRE-barrier-harms-landscape-appearance	Renewables undermine the appearance of the landscape	0.04	−0.86	0.00

(continued on next page)

Table B1 (continued)

No.	Question	Label	Variable-items	Skewness	Kurtosis	Kolmogorov-Smirnov p-value
19	q_120	CRE-barrier-undermines-fossil-industries	Renewables undermine our gas and coal businesses	−0.03	−0.78	0.00
Perceived Barriers to RE						
20	q_011	RE-barrier-fossil-fuel-sector	Fossil fuel sector creates RE obstacle	−0.19	−0.91	0.00
21	q_012	RE-barrier-large-businesses	Other large businesses create RE obstacle	−0.10	0.80	0.00
22	q_013	RE-barrier-grid-transmission	Grid transmission problems create RE obstacle	−0.21	−0.80	0.00
23	q_014	RE-barrier-local-government	Local Government policies, regulations or politics create RE obstacle	−0.08	−0.93	0.00
24	q_015	RE-barrier-state-government	State Government policies, regulations or politics create RE obstacle	−0.10	−0.82	0.00
25	q_016	RE-barrier-federal-government	Federal Government policies, regulations or politics creates RE obstacle	−0.11	−0.88	0.00
26	q_017	RE-barrier-community-opposition	Community opposition creates RE obstacle	−0.00	0.093	0.00
27	q_018	RE-barrier-financial-constraints	Financial constraints and considerations creates RE obstacle	−0.19	−0.82	0.00
28	q_020	RE-adoption-green-energy	I purchase renewable 'Green' energy through an electricity retailer (q_020)	0.67	−1.55	0.00
29	q_021	RE-adoption-installed-home	I have renewable energy installed (q_021)	0.14	−1.98	0.00
30	q_022	RE-adoption-home-battery	I have a battery for electricity storage (q_022)	0.83	−1.30	0.00
31	q_023	RE-adoption-electric-vehicle	I have an electric vehicle (q_023)	0.91	−1.17	0.00
32	q_024	RE-adoption	I plan to install renewables within the next 1–2 years	0.75	−1.43	0.00
33	q_025	RE-intent-physical-barriers	I want to generate more renewable energy but currently have barriers	0.82	−1.33	0.00
34	q_028	RE-satisfaction-existing-RE-EV	Satisfaction with installed solar or batteries	−0.43	−0.45	0.00
35	q_036	importance-blackout-protection	It is important for me to have blackout protection	−0.55	−0.61	0.00
36	q_029	experiences-blackouts-frequency	Are brownouts or blackouts a problem for you?	0.09	−0.74	0.00
37	q_030	experiences-grid-issues	Do you face any other issues from the electricity grid?	0.74	−1.45	0.00
Community Allegiance						
38	q_089	CRE-feeling-connected	I feel strongly connected to the community I live in	−0.99	0.97	0.00
39	q_090	CRE-feeling-positive	I think of my community as being a great place to live	−0.87	−0.53	0.00

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Table B1 (continued)

No.	Question	Label	Variable-items	Skewness	Kurtosis	Kolmogorov-Smirnov p-value
40	q_091	CRE-feeling-volunteer-participation	I volunteer in the community	−0.73	−0.63	0.00
41	q_092	CRE-feeling-division-on-fossil	There are big divisions in my community about coal or gas	−0.55	−1.06	0.00
42	q_093	CRE-feeling-charity-contributions	I make charitable contributions to the community	−0.82	−0.54	0.00
Willingness to Invest in RE/CRE						
43	q_034	RE-intent-install-solar-2yr	I intend to install a solar PV system within the next 1–2 years	−0.62	−0.41	0.00
44	q_035	RE-intent-install-other-2yr	I intend to install other renewable energy systems within the next 1–2 years	−0.49	−0.70	0.00
45	q_031	CRE-participation	Do you currently participate or have invested in a community energy group or project?	−0.31	0.38	0.00
Social Cultural Norms						
46	q_094	RE-social-network	Have any of your friends or neighbours got household renewable energy?	−0.85	−0.66	0.00
47	q_121	CRE-usage-pattern	If your energy bill decreases significantly through renewables, how might your energy usage change?	−0.28	−1.08	0.00
48	q_113	CRE-external-usage-management	Likelihood to participate in a program managing household energy remotely	−0.50	−0.89	0.00
Perceived Benefits of CRE						
49	q_037	CRE-motivation-social-connection	Community building/social connection	−0.21	−0.86	0.00
50	q_038	CRE-motivation-local-economy	Support local economy	−0.33	−0.79	0.00
51	q_039	CRE-motivation-financial-benefit	Personal financial benefit	−0.19	−0.86	0.00
52	q_040	CRE-motivation-peer-involvement	People I knew were involved	−0.16	−0.88	0.00
53	q_041	CRE-motivation-climate-action	Acting on climate change/carbon emissions reductions	−0.30	−0.91	0.00
54	q_042	CRE-motivation-energy-independence	Greater independence from commercial energy providers	−0.22	−0.95	0.00
55	q_043	CRE-motivation-decision-involvement	Being involved in decision-making	−0.22	−0.88	0.00
56	q_044	CRE-motivation-policy-support	Favorable government policies	−0.39	−0.67	0.00
57	q_099	CRE-benefit-jobs	Contribute well-paid, long-term jobs	0.79	−1.38	0.00
58	q_100	CRE-benefit-maintains-lifestyle	Help maintain my existing way of life	0.67	−1.55	0.00

(continued on next page)

Table B1 (continued)

No.	Question	Label	Variable-items	Skewness	Kurtosis	Kolmogorov-Smirnov p-value
59	q_101	CRE-benefit-protects-land-water	Reduce the risk of destruction of land and water	0.50	−1.75	0.00
60	q_102	CRE-benefit-improves-community-cohesion	Improve community spirit, cohesion, and resilience	0.89	−1.20	0.00
61	q_103	CRE-benefit-improves-external-image	Improve community image in the outside world	1.38	−0.09	0.00
62	q_104	CRE-benefit-keeps-money-local	Keep money in regional communities	1.63	0.66	0.00
63	q_105	CRE-benefit-supports-grants-sponsorships	Enable community sponsorship and grant opportunities	1.27	−0.39	0.00
64	q_106	CRE-benefit-supports-local-participation	Enable local participation in energy decision-making	0.78	−1.40	0.00
65	q_107	CRE-benefit-supports-businesses	Enable new businesses based on cheaper energy	0.81	−1.34	0.00
66	q_108	CRE-benefit-supports-local-networks	Make local networks (q_108)	1.45	0.10	0.00
67	q_109	CRE-benefit-increases-independence	Deliver independence from providers and control over the energy system	1.36	−0.14	0.00
Achievement of RE target						
68	q_000	RE-mix-preference	What is your preference for Australia's future energy mix as a percentage?	−0.30	−0.83	0.00
69	q_001	RE-target-preferred-year	When would you like to see your future energy mix target met?	−0.08	−1.09	0.00
70	q_003	RE-target-achieved-retrofit-buildings	Retrofitting buildings to be energy efficient	0.92	−1.16	0.00
71	q_004	RE-target-achieved-household-solar-batteries	Installing solar PV and/or batteries at household and business scale	0.40	−1.84	0.00
72	q_005	RE-target-achieved-community-scale	Small and medium scale community-owned renewable projects	0.58	−1.66	0.00
73	q_006	RE-target-achieved-corporate-scale	Medium and large scale corporate-owned renewables projects	0.75	−1.43	0.00
74	q_007	RE-target-achieved-electric-vehicle	Transition to electric vehicles	0.88	−1.22	0.00
75	q_008	RE-target-achieved-powered-manufacturing	Transition to renewable energy powered manufacturing	1.13	−0.72	0.00
76	q_009	RE-target-achieved-phasingout-fossil	Phasing out fossil fuel power generation	1.60	0.57	0.00
Demographics						
77	q_123	residential-area-type	Where do you live?	−0.43	−1.59	0.00
78	q_125	gender-identity	What is your gender?	0.26	−1.59	0.00
79	q_126	age-group	Which age range are you in?	1.98	5.54	0.00
80	q_130	state-of-residence	In which state do you live?	0.29	−1.34	0.00

(continued on next page)

Table B1 (continued)

No.	Question	Label	Variable-items	Skewness	Kurtosis	Kolmogorov-Smirnov p-value
Roles in CRE Projects						
81	q_045	CRE-role-investor	Investor	0.80	−1.34	0.00
82	q_046	CRE-role-employee	Employed or contracted to project	0.75	−1.40	0.00
83	q_047	CRE-role-volunteer	Volunteer for project	0.78	−1.12	0.00
84	q_048	CRE-role-decision-group	On decision-making group	0.72	−1.09	0.00
85	q_049	CRE-role-advisor	Provide expertise or on advisory group	0.70	−1.11	0.00
86	q_050	CRE-role-traditional-custodian	Traditional custodian advisor	0.65	−1.09	0.00
87	q_051	CRE-role-local-engagement	Talk to locals about the project	0.77	−1.13	0.00
88	q_052	CRE-role-policy-engagement	Talk to government or decision-makers	0.74	−1.12	0.00
89	q_053	CRE-role-land-provider	Provide or lease land/space to the project	0.68	−1.17	0.00
90	q_054	CRE-role-input-provider	Supply other inputs (e.g., crop waste for bioenergy)	0.66	−1.10	0.00

Table B2

Descriptive statistics for variables.

Variable	Mean	SD	Min	Max	Variable	Mean	SD	Min	Max
q_000	0.69	0.22	0.00	1.00	q_001	1.99	0.78	0.00	3.00
q_003	0.29	0.45	0.00	1.00	q_004	0.40	0.49	0.00	1.00
q_005	0.36	0.48	0.00	1.00	q_006	0.32	0.47	0.00	1.00
q_007	0.30	0.46	0.00	1.00	q_008	0.25	0.44	0.00	1.00
q_009	0.19	0.39	0.00	1.00	q_011	3.17	1.26	1.00	5.00
q_012	3.11	1.19	1.00	5.00	q_013	3.24	1.21	1.00	5.00
q_014	3.09	1.23	1.00	5.00	q_015	3.09	1.20	1.00	5.00
q_016	3.11	1.20	1.00	5.00	q_017	3.08	1.21	1.00	5.00
q_018	3.23	1.21	1.00	5.00	q_020	0.34	0.47	0.00	1.00
q_021	0.46	0.50	0.00	1.00	q_022	0.31	0.46	0.00	1.00
q_023	0.29	0.46	0.00	1.00	q_024	0.32	0.47	0.00	1.00
q_025	0.31	0.46	0.00	1.00	q_028	3.33	1.28	0.00	5.00
q_029	3.17	1.07	1.00	5.00	q_030	0.47	0.00	1.00	
q_031	0.99	0.10	0.00	1.00	q_034	3.29	1.37	0.00	5.00
q_035	3.25	1.40	0.00	5.00	q_036	3.39	1.30	0.00	5.00
q_037	3.20	1.20	1.00	5.00	q_038	3.28	1.20	1.00	5.00
q_039	3.14	1.20	1.00	5.00	q_040	3.10	1.20	1.00	5.00
q_041	3.45	1.10	1.00	5.00	q_042	3.55	1.15	1.00	5.00
q_043	3.35	1.00	1.00	5.00	q_044	3.20	1.05	1.00	5.00
q_045	0.50	0.50	0.00	1.00	q_046	0.30	0.46	0.00	1.00
q_047	0.25	0.43	0.00	1.00	q_048	0.22	0.42	0.00	1.00
q_049	0.50	0.50	0.00	1.00	q_050	0.40	0.49	0.00	1.00
q_051	0.33	0.47	0.00	1.00	q_052	0.29	0.46	0.00	1.00
q_053	0.31	0.46	0.00	1.00	q_054	0.28	0.45	0.00	1.00
q_080	3.25	1.25	1.00	5.00	q_081	3.18	1.18	1.00	5.00
q_082	3.22	1.20	1.00	5.00	q_083	3.17	1.17	1.00	5.00
q_084	3.12	1.15	1.00	5.00	q_085	3.10	1.14	1.00	5.00
q_086	3.08	1.13	1.00	5.00	q_087	3.05	1.10	1.00	5.00
q_088	3.00	1.05	1.00	5.00	q_089	2.95	1.00	1.00	5.00
q_090	2.90	0.98	1.00	5.00	q_091	2.85	0.96	1.00	5.00
q_092	2.80	0.95	1.00	5.00	q_093	2.75	0.93	1.00	5.00
q_094	2.70	0.92	1.00	5.00	q_099	4.10	0.94	1.00	5.00
q_100	3.35	1.12	1.00	5.00	q_101	4.02	0.99	1.00	5.00
q_102	3.90	1.01	1.00	5.00	q_103	3.84	1.03	1.00	5.00
q_110	3.56	0.98	1.00	5.00	q_104	3.11	1.22	1.00	5.00
q_105	3.13	1.23	1.00	5.00	q_106	3.15	1.24	1.00	5.00
q_107	3.17	1.25	1.00	5.00	q_108	3.19	1.26	1.00	5.00
q_109	3.21	1.27	1.00	5.00	q_110	3.56	0.98	1.00	5.00
q_111	3.59	0.94	1.00	5.00	q_112	3.60	0.96	1.00	5.00
q_113	3.41	1.32	1.00	5.00	q_114	3.16	1.16	1.00	5.00
q_115	3.12	1.11	1.00	5.00	q_116	3.34	1.03	1.00	5.00
q_117	3.49	1.09	1.00	5.00	q_118	3.25	1.13	1.00	5.00
q_119	3.19	1.11	1.00	5.00	q_120	3.26	1.09	1.00	5.00
q_121	2.04	0.85	0.00	3.00					

Appendix C. SHAP-based methods for explaining ML models

C.1. Shapley value computation

Shapley values are a concept from coalitional game theory that provide a way to fairly distribute the payout of a game among its players. In the context of machine learning, the feature values of a data instance act as players in a coalition, and the “payout” is the prediction made by the model. Shapley values ensure a fair distribution of the prediction value among all input features by considering all possible coalitions of features. In our context, the Shapley value ϕ_i for response to the i -th survey question q_i is defined in Eq. (C.1):

$$\phi_i = \sum_{S \subseteq Q \setminus \{q_i\}} \frac{|S|! \cdot (|Q| - |S| - 1)!}{|Q|!} \left(f_{S \cup \{q_i\}}(x_{S \cup \{q_i\}}) - f_S(x_S) \right) \quad (C.1)$$

where S is a subset of all survey questions Q excluding q_i , $f_{S \cup \{q_i\}}$ is the model trained with questions in S and q_i , and x_S are the responses to questions in S . This Shapley value ϕ_i represents the average contribution of the i -th survey response to the prediction of the level of participation across all possible combinations of survey questions.

C.2. Tree SHAP explainer

Tree SHAP adapts the Shapley value approach specifically for tree-based ML models such as random forests and gradient-boosting trees. Shapley values are calculated by adding each feature sequentially into a conditional expectation function that estimates how tree model’s output would change upon the addition. The effects of the additions are then averaged across all possible feature orderings (Lundberg et al., 2020). Formally, Tree SHAP computes SHAP values recursively by traversing the tree, with Eq. (C.2):

$$\phi_i(f, x) = \sum_{R \in \mathcal{R}} \frac{1}{M!} (f_x(P_i^R \cup \{i\}) - f_x(P_i^R)) \quad (C.2)$$

where f_x is the conditional expectation function, $\mathcal{R}$ denotes the set of orderings of all survey questions, P_i^R represents the collection of questions that precede question i in a particular ordering R , and M indicates the total count of questions included in the tree model.

C.3. Kernel SHAP explainer

Kernel SHAP integrates the concept of LIME (Ribeiro et al., 2016), which explains predictions of any classifier by approximating it locally with an interpretable model, with Shapley values to develop a model-agnostic method for estimating feature contributions. It uses an additive feature attribution method (Lundberg and Lee, 2017), where the explanation model g is a linear function of binary variables representing the presence or absence of survey responses in Eq. (C.3):

$$g(z') = \phi_0 + \sum_{i=1}^N \phi_i z'_i, \quad (C.3)$$

where $z' \in \{0, 1\}^N$ denotes the absence or presence of question i , N is the number of survey questions, and ϕ_i are the question attributions. To estimate the Shapley values, Kernel SHAP uses a specific form of weighted linear regression. This is achieved by solving the following optimisation problem in Eq. (C.4):

$$L(f, g, \pi_{x'}) = \sum_{z' \in \mathcal{Z}} [f(h_x(z')) - g(z')]^2 \pi_{x'}(z'), \quad (C.4)$$

where $\pi_{x'}$ is a weighting kernel designed to match the Shapley values. That is, the loss function evaluates how well the linear model approximates the real model’s predictions across all coalitions, weighted by their respective $\pi_{x'}$.

C.4. Deep SHAP explainer

Deep SHAP combines DeepLIFT (Shrikumar et al., 2016) with Shapley values to create a method specifically tailored for deep learning models such as neural networks. DeepLIFT attributes a value to each input feature based on the difference between its actual value and a reference value, propagating these differences through the layers of the network, with Eq. (C.5):

$$\sum_{i=1}^n C_{\Delta x_i \Delta o} = \Delta o, \quad (C.5)$$

where $C_{\Delta x_i \Delta o}$ represents the contribution of input x_i to the change in model output $o = f(x)$, $\Delta o = f(x) - f(r)$, $\Delta x_i = x_i - r_i$, and r denotes the reference input. In our survey context, x_i represents the response to the i -th survey question, and o represents the model’s prediction of the participation level. Deep SHAP approximates SHAP values by interpreting DeepLIFT’s reference value as the expected response of the input questions. It uses a linear composition rule to combine SHAP values computed for smaller components of the network (Lundberg and Lee, 2017). Deep SHAP recursively combines SHAP values for individual layers or components of the network, leveraging the compositional nature of deep networks.

Appendix D. Supporting analyses

D.1. SCI robustness assessment: Empirical validation

We provide empirical validation of SCI through two comparisons with standard Pearson correlation and decomposed averaging.

Table D1 compares standard Pearson correlation (aggregating across all classes) against SCI computed separately per participation level. At 5% noise to influential features, standard Pearson favors Kernel SHAP (0.976) over Tree SHAP (0.968). However, SCI reveals Tree SHAP maintains uniform high stability across all participation levels (Low = 0.999, Medium = 0.999, High = 0.998), while Kernel SHAP shows lower stability (0.956, 0.973, 0.972). For policy applications targeting Medium participation—threshold citizens requiring institutional trust interventions—Tree SHAP provides superior stability (99.9% vs. 97.3%).

Similarly, Deep SHAP’s standard Pearson (0.512) appears marginally acceptable, yet SCI reveals near-zero or negative correlations (0.020, 0.013, −0.002)—severe instability masked by global aggregation. At 2% noise, SCI detects Medium and High class failures (negative correlations: −0.005, −0.022) while standard Pearson suggests acceptable stability (0.687). This pattern demonstrates that standard Pearson can favor suboptimal explainers by masking class-level instabilities through aggregation.

Table D2 compares SCI’s joint evaluation of the full instances×features matrix against decomposed approaches that average correlations separately across features or instances. For Deep SHAP at 5% noise,

Table D1
Standard Pearson correlation vs. SCI across noise levels.

Explainer	Noise	Standard Pearson	SCI		
			Low	Medium	High
Tree SHAP	2%	0.985	1.000	0.999	0.999
Kernel SHAP	2%	0.991	0.956	0.974	0.973
Deep SHAP	2%	0.687	0.010	−0.005	−0.022
Tree SHAP	5%	0.968	0.999	0.999	0.998
Kernel SHAP	5%	0.976	0.956	0.973	0.972
Deep SHAP	5%	0.512	0.020	0.013	−0.002
Tree SHAP	10%	0.951	0.998	0.999	0.997
Kernel SHAP	10%	0.963	0.953	0.971	0.971
Deep SHAP	10%	0.378	0.012	−0.008	0.012
Tree SHAP	20%	0.912	0.997	0.995	0.986
Kernel SHAP	20%	0.934	0.943	0.959	0.959
Deep SHAP	20%	0.201	0.037	0.033	0.043

Table D2
Decomposed averaging vs. joint N×M evaluation.

Explainer (Class)	Noise	Feature-wise Avg Corr	Instance-wise Avg Corr	SCI (Joint N×M)
<i>Deep SHAP — Averaging Masks Instability:</i>				
Low class	2%	0.423	0.456	0.010
Medium class	2%	0.389	0.421	−0.005
High class	2%	0.412	0.445	−0.022
Low class	5%	0.356	0.389	0.020
Medium class	5%	0.312	0.342	0.013
High class	5%	0.378	0.412	−0.002
Low class	10%	0.289	0.312	0.012
Medium class	10%	0.245	0.267	−0.008
High class	10%	0.301	0.334	0.012
<i>Tree SHAP — All Approaches Converge:</i>				
Low class	5%	0.996	0.997	0.999
Medium class	5%	0.995	0.996	0.999
High class	5%	0.987	0.989	0.998
<i>Kernel SHAP — Minimal Divergence:</i>				
Low class	5%	0.961	0.964	0.956
Medium class	5%	0.976	0.979	0.973
High class	5%	0.965	0.968	0.972

feature-wise averaging suggests 31–38% stability and instance-wise averaging suggests 34–41% stability, yet SCI reveals near-zero or negative correlations (0.020, 0.013, −0.002). At 2% noise, the divergence is even more pronounced: decomposed approaches suggest 39–46% stability while SCI reveals severe instability including negative correlations for Medium (−0.005) and High (−0.022) classes.

This divergence occurs because averaging masks instability through aggregation. A feature may maintain average importance across the population even when individual attributions shift dramatically—for example, policy risk (q₁₁₂) might maintain mean SHAP value ≈0.15 across instances while individual respondents' attributions change unpredictably under perturbation. Similarly, an individual's attribution profile may appear stable in aggregate even when specific feature contributions change completely—which features receive high attribution can shift entirely while averaged magnitudes remain similar. SCI evaluates whether the full explanation surface—which features are important for which instances—remains stable, capturing instabilities that decomposed approaches miss.

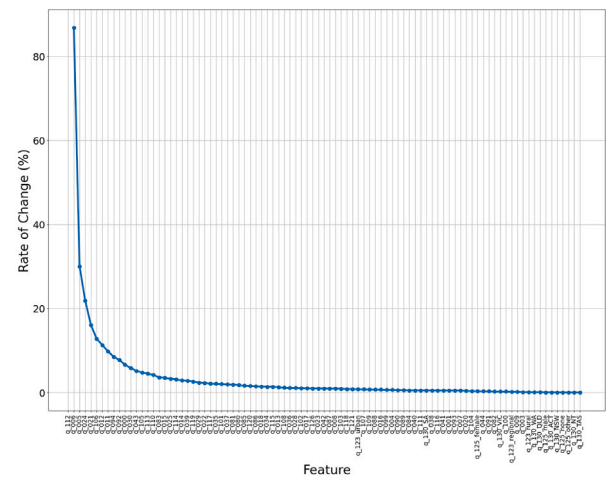
For Tree SHAP, all approaches converge (all values > 0.98), confirming SCI does not create false alarms when stability is genuine. For Kernel SHAP, minimal divergence occurs (decomposed: 0.961–0.979, SCI: 0.956–0.973), demonstrating moderate stability across evaluation methods.

D.2. Intersecting features across all models and explainers

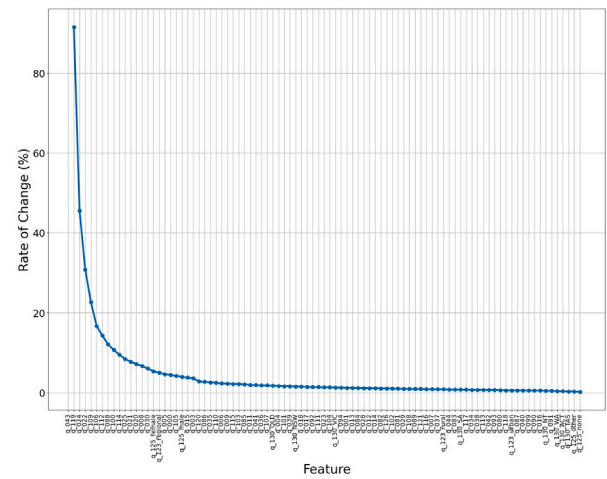
To establish a criterion for significance, we initially examined the rate of change in the cumulative feature contribution for each explainer, as illustrated in Fig. D1.

To enhance global explainability, we concentrated on identifying a model-agnostic feature space, aiming to develop insights less influenced by specific model architectures and more reflective of the underlying question responses and their characteristics. Echoing the analysis described in Section 4.2, we applied a significant threshold of 3% to delineate the class-wise important features for the Tree, Kernel, and Deep explainers. By investigating the intersections of these features, we identified core sets that consistently influence model decisions across various participation levels, underscoring their pivotal role in shaping participation decisions. The results are presented as follows:

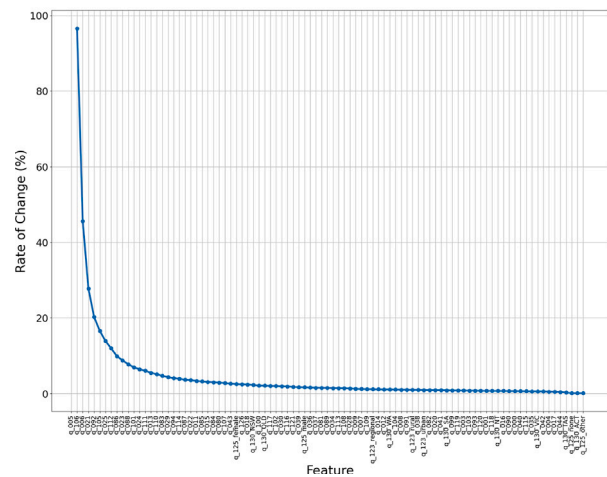
Low : [q₀₀₅, q₀₂₁, q₀₂₂, q₀₂₄, q₀₂₅, q₁₀₅, q₁₀₆, q₁₁₂]
Medium : [q₀₁₅, q₀₄₃, q₁₁₂, q₁₁₄]
High : [q₀₀₅, q₀₀₆, q₀₂₁, q₀₂₄, q₀₂₅, q₀₉₂, q₁₀₆]



(a) Tree SHAP



(b) Kernel SHAP



(c) Deep SHAP

Fig. D1. Determining important features based on rates of change in cumulative contributions.

Features such as q₀₀₅ and q₀₀₆ gauge community preference between small & medium scale community-owned and medium & large scale corporate-owned renewable projects, respectively. The likelihood of state government policies, regulations, or politics obstructing RE goals

is assessed in q_015. Personal investment in renewable technologies is probed with q_021 and q_022, where respondents disclose whether they have installations like solar PV or wind turbines and if they possess energy storage solutions, respectively. Future intentions toward adopting renewable technologies are explored through q_024, while q_025 identifies logistical challenges that might inhibit such installations, such as housing conditions or geographical constraints. The importance of participatory decision-making in CRE projects is queried in q_043, reflecting community interest in governance aspects of CRE projects. Community divisions concerning traditional energy sources like coal or gas are examined in q_092. Perceived benefits of RE, such as community sponsorship opportunities and enhancing local decision-making, are explored through q_105 and q_106.

D.3. Local explanation provided by XGBoost-tree explainer

The analysis employs force plots to provide localized interpretations of XGBoost-Tree Explainer model for CRE level of participation. Single-instance force plots illustrate how individual features influence the prediction for a specific instance, as depicted in Fig. D2. According to (Lundberg et al., 2020), these plots offer a direct visualization of the contribution of each feature - where red bars indicate features that drive the prediction toward a particular class (a specific participation level in our case) and blue bars indicate features that pull the prediction away.

In a representative example, the model predicts a medium participation level by producing the highest log-odds of 1.93 for Medium, with q_043 = 1 being the most positive factor and q_112 = 3 being the most

negative. This means for this participant, considering 'being involved in decision making is not important' strongly contributes to medium participation rather than high participation (q_043 = 1 is blue for High). Feeling neutral about 'joining a community energy project is risky' diminishes the likelihood of a medium participation, and instead shifts the prediction to a high participation (q_112 = 3 is red for High).

To expand upon individual instance analysis, stacked force plots were also generated for the entire testing set, as shown in Fig. D3. This approach allows us to observe how features differentially influence the model confidence in its predictions across a broader array of cases. From the force plots, we observe that the model assigns similar levels of confidence to the low and medium participation predictions, while predictions for high participation are less frequent. This is evidenced by the smaller positive feature influences (red regions) for the 'High' class. This behavior aligns with the fact that high participation is the minority class in the dataset. Given that force plots in Fig. D3 aggregate all participant instances, they allow for the efficient examination of the directional influence of question responses on specific instances of interest. Consider, for example, the case of identifying drivers for high participation: instance 83 emerges as a notable example, exhibiting the highest log-odds at 1.698. The most influential responses contributing to this decision include 'q_006 = 1' (support for achieving RE targets through medium and large-scale corporate-owned projects), 'q_112 = 2' (disagreement that joining a community energy project is risky due to potential governmental policy changes) and 'q_024 = 1' (planning to install renewables).

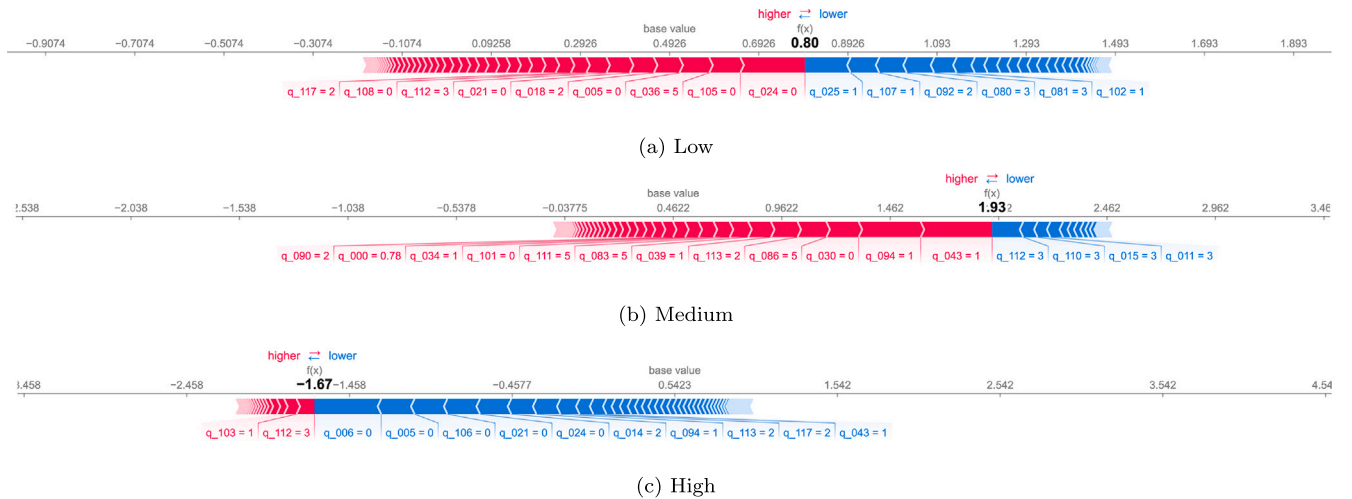


Fig. D2. Class-wise force plots of one instance.

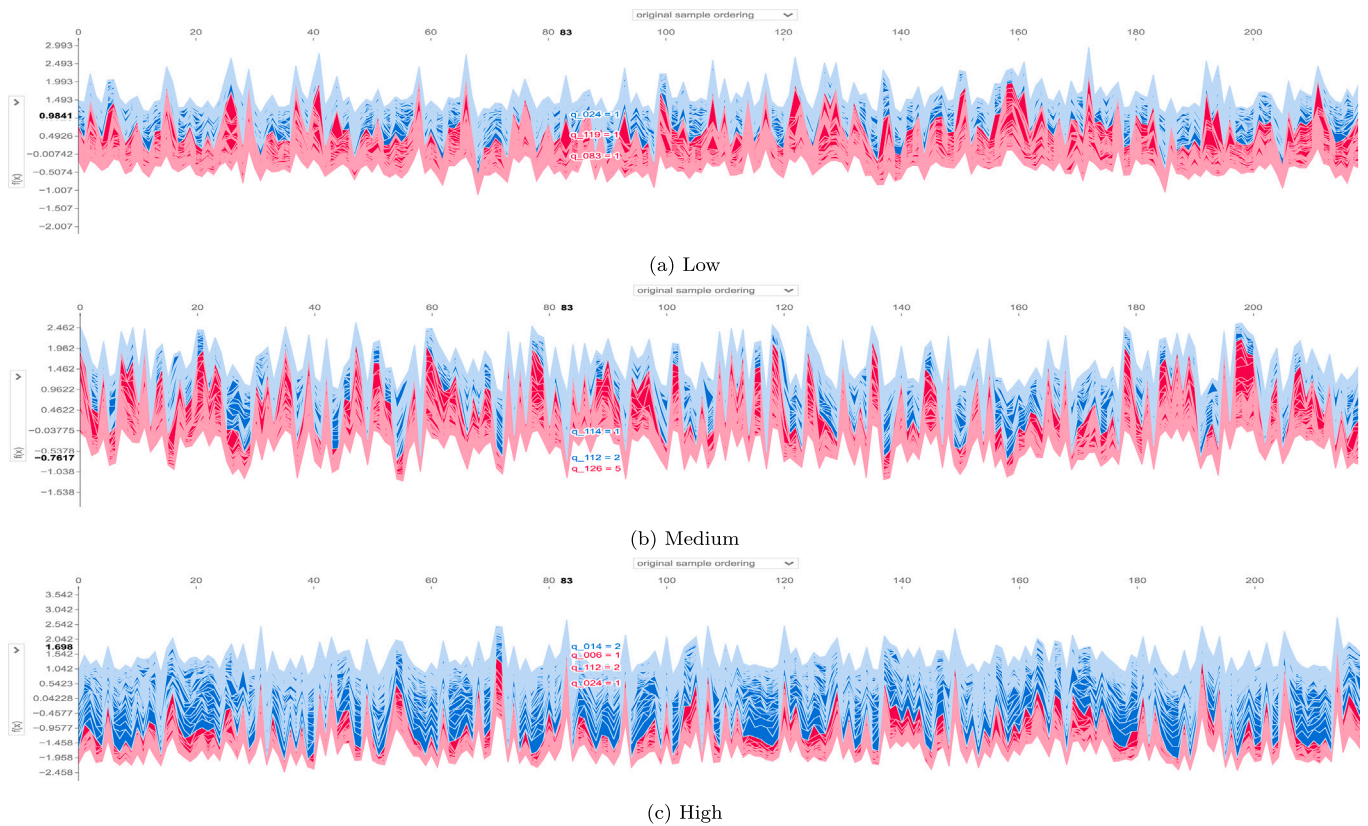


Fig. D3. Class-wise force plots of all instances.

Appendix E. Individual policy recommendations by participation level

Table E1 outlines a set of targeted policy recommendations derived from the SHAP-based feature analysis for each participation tier. Rather than presenting broad bundles, this table isolates individual predictors and links them to specific, actionable interventions. Each policy is aligned with the most affected participation level (low, medium, or high), enabling precision-targeted efforts. For example, constraints like energy storage access (q_{022}) and limited awareness of community funding (q_{105}) are linked to low participation, suggesting they are critical barriers to early engagement. In contrast, policy trust issues (q_{112}) and perceived decision-making influence (q_{043}) appear more relevant for advancing medium to high involvement. This granularity supports modular policy design, allowing practitioners to tailor strategies to specific community segments or obstacles.

The *Community Engagement and Governance* bundle addresses the political and relational dimensions of participation. The perceived effectiveness of small/medium community-owned projects (q_{005}) showed SHAP values of 0.10–0.25 for high participation, indicating that confidence in community-scale solutions drives deeper engagement. Notably, community sponsorship opportunities (q_{105}) exhibited asymmetric effects: positive SHAP contributions for low participation but near-zero or negative effects for high participation, suggesting that grant-based framing attracts initial interest but does not sustain advanced engagement. This finding implies that Australian community energy programs—such as those administered through the Clean Energy Finance Corporation—should evolve messaging from “funding opportunities” toward “ownership and governance participation” as communities mature.

The *Institutional Trust and Stability* bundle responds to our most striking asymmetric finding: policy risk perception (q_{112}) emerged as a dominant predictor specifically for medium participation (SHAP values 0.15–0.25) but showed minimal influence on high participation. This asymmetry suggests that regulatory uncertainty disproportionately affects citizens on the threshold of deeper engagement—precisely those whom policy should target for conversion. State government policy concerns (q_{015}) similarly predicted medium engagement (SHAP 0.08–0.15), reflecting Australia’s federal-state policy fragmentation. Corporate RE effectiveness perceptions (q_{006}) demonstrated the strongest class-specific effect in our analysis, with SHAP values of 0.20–0.60 for high participation only—suggesting that confidence in large-scale corporate projects complements rather than substitutes for community engagement among highly active citizens. Policy mechanisms such as grandfathering provisions, longer regulatory review cycles, and bipartisan renewable energy commitments could address these trust barriers.

The *Social Norms and Peer Influence* bundle leverages the normative factors our model identified. Peer RE adoption (q_{094}) showed moderate positive SHAP values (0.05–0.12) across participation levels, confirming social proof effects documented in prior research. Community divisions about coal or gas (q_{092}) emerged as a contextual moderator, with higher values associated with both very low and very high participation—suggesting that energy conflict can either suppress engagement (in divided communities) or motivate activism (among committed advocates). Australian programs could leverage these dynamics through neighborhood-level “solar champions” initiatives, building on successful models from the Australian Energy Foundation’s community engagement programs.

Table E1

Targeted policy recommendations by participation level.

Factor	Recommended policy	Shift
q.024 – Intention to install renewables	Launch subsidy or zero-interest financing programs to reduce up-front costs for installing RE at home.	Low → High
q.005 – Community-scale projects	Fund visible, successful pilot CREs in low-participation areas to build trust and local relevance.	Low → High
q.006 – Corporate-scale RE projects	Communicate benefits of corporate RE (e.g., job creation, stability) clearly to communities.	Low → High
q.021 – Existing home RE installations	Offer technical assistance and incentives for solar/battery installation.	Low → High
q.106 – Local participation in decisions	Establish neighborhood energy planning committees to increase civic engagement.	Low → High
q.025 – Barriers to adoption	Create rental/tenant-friendly RE options such as solar leasing or shared community assets.	Low → High
q.022 – Energy storage access	Subsidize battery storage systems or introduce shared storage solutions in communities.	Low → Medium
q.092 – Community division	Organize local deliberative forums to bridge views and resolve energy-related tensions.	Low → Medium
q.105 – Community grants	Expand awareness and simplify access to funding for local CRE initiatives.	Low → Medium
q.043 – Decision-making involvement	Involve locals in real decisions via community advisory boards or steering groups.	Medium → High
q.112 – Policy risk concerns	Promote stable, cross-party RE policy commitments to reduce fear of shifting rules.	Medium → High
q.114 – Perceptions of RE availability	Educate on storage/hybrid systems that ensure 24/7 RE access.	Medium → High
q.015 – State policy concerns	Clarify state-level RE policies and communicate long-term roadmaps with consistency.	Medium → High

Data availability

Data will be made available on request.

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